INTRODUCTION: Animal breeding Programmes- an overview

Animal breeding is a fascinating discipline. It is entirely logical and can be very abstract. For example, you can never touch a breeding value!

The underlying principles of animal breeding are the same for all livestock species, only the breeding technologies applied differ.

Animal breeding programmes exist within a particular political, economic, or physical climate over which breeders have little or no control.

--eg, breeders have no control over:

- Markets for selling animal products.
- Government subsidies
- Trade restrictions
- Price controls
- Altitude, rainfall
- Growing seasons
- Consumer attitudes etc

Before establishing any breeding programme, two basic questions need to be answered.

What is the best animal?

How do you breed animals so that their descendants will be at least better than today's animals?

To answer the questions, one need to know:

- ***Traits of economic importance for each species***
- ***What phenotypes and genotypes are most desirable for those traits***

To develop a sense of the importance of traits and best genotypes in a more objective way, one must understand that the genotype of an animal is just one part of a much larger **system**.

Definition: System-- a group of interdependent component parts.

Animal related systems vary from very large extremely complex industries to small farms. For example, the components of a system such as a farm:

- ***Animals (genotypes)*** -- one or many e.g. Dairy farm

- ***Physical environment*** -- elements of the environment over which humans exert little control (weather, altitude, soils etc)-- particularly important when considering range cattle production or pastoralism
- ***Fixed resources*** -- include the size of the farm, infrastructure, available labour etc.
- ***Management*** -- involves all the policies implemented by the farmer eg level of supplementary feeding, health care and length of time animals remain on the farm
- ***Economics*** -- refers to the costs of farm inputs and the prices of farm outputs. Also long and short term interest rates.

Note:

- All the components within a system interact with each other. For example, the best preventive health care programme (management) depends on the kind of pathogens present (physical environment), and the costs of vaccines, dewormers and drugs (economics)
- To determine what is best, or most cost-effective, one must have knowledge of alternative programmes!

For many species, ***Genotype X Environment*** interactions play a critical role in determining the most appropriate biological type for a given environment. Recall: G X E Interaction occurs when the ***relative*** performance of different genotypes ***depends*** on the environment they are raised in. eg. Holstein Friesian Cattle in Temperate vs Tropical environments

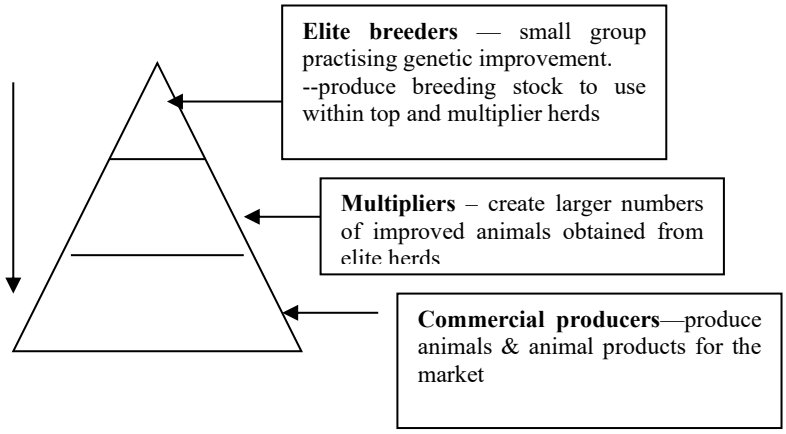
An understanding of the entire system and the many interactions among components of the system is thus the Key to determining the optimal genotype for an environment.

In trying to determine the best animal, a question of interest is: Best for whom?

- This depends on the structure of the breeding industry and the place of the breeder in that structure.

Breeding structures

Breed improvement generally follows a pyramid structure:



In poultry and pig production, large breeding companies take up the top two tiers of the structure –eg. Kenchick hatcheries, Farmer's choice

In the sheep and beef industry, the pyramid structure is more evident

In the dairy industry, the multiplier level doesn't really exist because the widespread use of Artificial insemination (AI) means that elite sires are available to all levels of producers.

Generally, genetic progress is made in the top tier and is passed down to the commercial producers. There is a large time lag in progress between the various tiers.

In the dairy industry, decisions made today will affect the population ten years down the line, whereas in poultry and pig production, the time taken to see the effects of a breeding decision is much less.

The pyramid suggests a flow of germplasm (genetic material eg live animals, semen, embryo's) from the top down.

Ideally, breeders at each level try to produce animals that will be in greatest demand by the customers at the next level down.

 Breeders at all levels must tailor their **Breeding Objectives** to meet the demands of the end users.

Definition: Breeding objective-- a general goal for a breeding programme-- a notion of what constitutes the best animal.

For any practical animal breeder, defining the breeding objective is one of the first steps to be taken.

Running an ***animal breeding program*** involves answering questions, such as:

1. What is the breeding objective? Which traits need to be improved and how important are different traits in relation to each other?

2. What and who is to be measured? (Which traits, which animals)

3. Do we need to use any reproductive technology (Artificial Insemination, Embryo Transfer)?

4. How many and which animals do we need to select as parents for the next generation?

5. How do we mate the selected males and females?

Note:

- Many practical breeding programs suffer from the fact that the **objectives are not properly defined**.
- The outcome for breeding programs is noticed many years after selection decisions are made. Hence objectives have to be designed for future circumstance. It is quite difficult to predict such circumstances, and it is even harder to define objectives that are reasonably stable over time.

Breeding strategies

Once a producer has determined the objective, the next issue is: ***How to get there***? The purpose of animal breeding is not to genetically improve individual animals --once an individual is conceived, there is not much one can do to change him!-- but to improve animal populations, to improve future generations of animals.

To try and determine How--breeders have two basic tools:

- **Selection**: the process that determines which individuals become parents, how many offspring they may produce, and how long they remain in the breeding population.
- **Mating**: decision on which of the selected males will be bred to which females that have been selected.

Breeding strategies involve using these two tools.

The idea behind selection is to let the individuals with the best sets of genes reproduce so that the next generation has on average more desirable genes than the current generation--ie chose animals with the best ***Breeding Values***.

- The Reproductive rate of breeding animals and uncertainty about true genetic merit of breeding animals make up the most important limiting factors in a breeding programme.
- Investments in breeding programs are therefore often related to trait measurement and genetic evaluation, and to technology to increase reproductive rates.

Basic components of a breeding strategy

a) Determine a breeding objective/ goal

b) Maintain livestock records

National livestock databases can only be developed through animal recording schemes.

Successful animal recording cannot be viewed as limited only to the high input production systems. Farmers at all levels must be involved, then appropriate information can be collected and results can be interpreted and acted on.

What is animal recording?

Animal recording involves the measurement of various indicators of animal performance and the use of that information in the decision making process. Key elements of the process of recording include:

- Measurement of performance
- Recording of the data

- Processing of the data to extract the desired information
- Use of the information and data to make management decisions

Measurement and recording of data

Measurements should be objective and accurate -- not complex or require sophisticated equipment.

Eg record dates of parturition, record weights on specific dates

If farmers are involved in the measurement and recording, they will readily experience the benefits.

The benefit of abundant and good measurement is that we may better be able to identify the genetically superior animals. This leads to more accurate selection and more genetic improvement.

Reasons for keeping particular records should be clear.

Only records that are important and applicable should be kept (with experience and farmer enthusiasm more detailed data can be collected with time).

If resources are limited, trait measurement may be confined to a small group of animals, thus creating an 'elite group (nucleus) of animals that will most likely be chosen as future parents.

Use of collected data

The most important use of data is in ***making management decisions***. Hence what is required is:

- An understanding of the value and utility of the measurements taken
- Timely and affordable processing of the data to make it understandable and useful
- Technical support at farm level to assist in interpretation of the processed data
- Feedback mechanisms to document the benefits gained from use of the data

Measurement effort is an investment, and the cost-benefit of it should be clear. The decision about which traits are good to measure can best be worked out in a selection index context (see later).

In some instances, like in dairy cattle, recording is done for more reasons than just genetic improvement alone. When recording is an investment for genetic improvement, a simple cost-benefit analysis could help in deciding which animals should be measured.

Phenotypic measurements are turned into *Estimated Breeding value's (EBVs)*. Nowadays, sophisticated statistical methods are used, leading to *Best Linear Unbiased Prediction* (BLUP) of breeding values. A genetic evaluation system using BLUP relies on good data measurement, a good structure of data (breeding animals across herds) and proper pedigree recording. If these prerequisites are in place, investment in BLUP methodology is usually highly cost efficient.

c) Carry out Genetic evaluation and selection

The decision about which animals should be selected as parents for the next generation is mainly based on *assessment of breeding value* of individual animals.

Genetic evaluation is the activity that deals with this assessment. It is central to animal improvement schemes.

- Estimation of breeding value needs to be based on trait measurement, either on the animal, or on animals genetically related to the individual (parents, sibs, or progeny).
- Estimation of breeding value based on an animal's phenotype alone can be quite accurate for high heritable traits.
- A progeny test is usually the most accurate assessment of breeding value, given that sufficient progeny are tested.

The ***accuracy of the breeding value*** is directly related to ***selection efficiency*** and ***rate of genetic improvement***.

Selection accuracy is strongly dependent on the degree of data recording.

If ***BLUP*** is used to generate EBVs also an ***animal's pedigree*** needs to be known (in principle, for each animal only sire and dam). If pedigree is not recorded, breeding value can be assessed on own performance only, and is limited to sexes, which express the traits of interest.

Selecting animals based on estimated breeding value maximizes the response to selection that can be achieved.

However, when deciding which animals should have offspring, the ***common ancestry*** *(*coancestry) of all selected parents should stay below certain limits, since it is directly related to the build up of ***inbreeding***. Coancestry among selected parents is determined by the average relationship among the selected parents as well as the number of parents selected.

d) Mating

Decisions about which animals need to be mated are often seen in relation ***to dominance effects***.

Note: Utilizing dominance variation is often not of primary importance for improvement of purebreds, but it can have more impact if breeding animals are selected from different breeds or lines, as heterotic effects between breeds can be utilized.

When ***multiple traits are involved in the breeding objective***, **assortative mating** could be useful, matching qualities in different parents for different traits. Formally, this is only useful if the traits show optimum values. Otherwise, improvement of traits is linear, and a certain disadvantage for trait A could be offset by increased advantage for trait B.

Decide on what Reproductive technology to use

Use of reproductive technology can be another *investment option.* ***Reproductive technology allows the intensive use of superior breeding stock***

Most of the main factors that determine genetic gain, are directly influenced by the reproductive rate of the breeding animals:

- A higher reproductive rate leads to the need for a decreased number of breeding animals, therefore increasing the intensity of selection of these animals.
- If reproductive technology is possible, for example AI, the benefit could be expressed in terms of increased genetic rate of improvement, which in turn has a monetary component attached to it. More offspring per breeding animal allow also more accurate estimation of breeding value.
- As new technologies in animal breeding allow faster genetic change, long term issues such as ***inbreeding and maintenance of genetic variation*** become important.

As reproductive rates are basically multiplying factors in a breeding structure, any improvement in reproduction will justify higher investment in improvement of the best breeding stock. Moreover, the reproductive rates have a large impact on socio- economic structures of a livestock industry. It is not a coincidence that pigs and poultry are owned by only a few breeding companies worldwide, often with a major interest of multinational companies. On the other hand, beef cattle breeding, where natural mating is predominately practiced, is typically in the hands of a large number of breeding studs.

Animal Genetic resources

Genetic diversity, species, breeds, strains and populations
Diversity: Generally refers to the range of differences among some set of entities

Biological diversity: refers to variety within the living world, the biosphere.

Biodiversity: is commonly used to describe the number, variety and variability of living organisms.

Diversity can be measured if some quantitative value can be ascribed to it, and the values compared. In order to do this, ***biodiversity*** is divided into its constituent elements: Genes, species and ecosystems.

The discussion on global diversity is typically presented in terms of global numbers of ***species*** in different taxonomic groups. Total number of species existing on earth at present varies from 5-100 million! Of these, only 1.7 million have been described to date. Using species alone is not adequate to describe diversity!

Genetic diversity represents the heritable variation within and between populations of organisms. The population may be an entire species or a specific collection of individuals within a species such as a breed, strain, line, herd/flock. The diversity ultimately lies in the genetic code (determined by nucleic acids).

New genetic variation arises in individuals by gene and chromosome mutations, and is spread through the population by recombination.

Animal genetic resources (AnGR)

The term AnGR is used to include all animal species, breeds and strains that are of economic, scientific and cultural interest to humankind in terms of food and agricultural production for the present and the future.

Another equivalent, increasingly used term is **farm animal genetic resources**. There are more than 40 species of animals that have been domesticated (or semi-domesticated) during the past 10-12 thousand years, which ***contribute directly***--through animal products used for food and fibre; and ***indirectly--*** through functions and products such as draft power, manure, transport etc.

Common species include cattle, sheep, goats, pigs, chicken, horses, buffalo etc and others (ref table!)

Domestic animal Diversity (DAD)

Refers to the genetic variation or diversity existing among the species, breeds, strains and individuals for all animal species which have been domesticated, and their immediate wild relatives.

The concept of a ***breed***, in which all members have a pedigree tracing their ancestry, was developed primarily in Western Europe during the eighteenth century.

Livestock populations developed in different ecological or geographic areas will become genetically distinct as a result of ***genetic drift and differential selection pressures***, provided they have also been ***reproductively isolated*** from other populations developed under different conditions.

There is an FAO database on AnGR -- ***DAD-IS***

This seeks to document all information on various strains / breeds found in different countries.

There is also a ***world watch list for domestic animal diversity (WWL-DAF)*** prepared by the FAO of the UN. First edition in 1993., revised in 1995 and 2000, and is to be revised again to incorporate info from developing countries.

A relatively small proportion of the world's livestock are listed in herd-books, so documentation on all breeds/ strains of livestock available is limited.

As breeders, work must be done at two levels:

- where there are populations that are clearly distinct and well documented
- where populations are not so easily discriminated, and documented information is lacking. Here genetic information is imperfect, and cultural information /definition of differences need to be considered.

It is a challenge thus to try and document info on AnGR. Current work being carried our by organizations eg ILRI.

Characterization

The distillation of knowledge which contributes to the reliable prediction of genetic performance of an AnGR in a defined environment and provides a basis for distinguishing between different AnGR and for assessing available diversity/

Characterization should include:

- A clear definition of the genetic attributes of an AnGR (indications of performance levels—milk, meat, growth. Reproduction etc)
- The environment to which it is adapted/ predominantly found
- Population size of the AnGR
- Its physical description – characters which facilitate identification
- Uses
- Prevalent breeding systems
- Population trends
- Predominant production systems

A thorough review of, and synthesis of data from both conventional and "grey" literature should be a first step in any breed characterization work.

Surveys, including farmers—to determine the extent of "indigenous" knowledge on the breed should also be carried out.

Attempting to obtain breed level statistics over a whole country is not practical. Rather , use of sampling techniques is more useful.

Mating Systems

Mating systems can be defined as a set of rules for making mating decisions. There is no limit to the possible number of mating systems. There are however only a few mating strategies. These include:

Random Mating –a system of mating in which mates are chosen at random. Usually no performance records or genetic predictions are required for this type of mating system, and little time is involved in making the decisions.

Assortative mating –of which there are two forms:

Positive assortative mating—the mating of similar individuals

Negative assortative mating—the mating of dissimilar individuals

Any mating strategy that is not random is generally a form of assortative mating.
Assortative mating requires
--performance records, genetic predictions or some other mating criteria.
--Animals are ranked in some order and multiple traits are usually considered.

SELECTION PROGRAMMES ACROSS BREEDS - CROSSBREEDING

Crossbreeding can be defined as the mating of sires of one breed or breed combination to dams of another breed or breed combination.

Line-crossing
can be defined as the mating of sires of one line or line combination to dams of another line or line combination. [A line is a group of related animals within a breed]
Breeds or lines within breeds can be randomly crossed, or assortatively mated.
In practise, crossbreeding programmes are rarely random.

Genetic basis of crossbreeding
The genetic basis of crossbreeding can broadly be classified into two: additive and nonadditive.

1. *Additive component*. This is that which is due to the averaging of merit in the parental lines or breeds, with simple weighting according to level of gene representation of each parental breed in the crossbred genotype.

- Additive genetic effects originate from contributions of single loci and are therefore transferred directly from parent to progeny.



This additive component can be divided into the

- ***individual additive genetic effects***
- ***maternal additive genetic effects***

- The individual additive genetic effect is the contribution to offspring phenotype attributable to its own set of genes.
- Maternal additive genetic effects are defined as any contribution or influence on the offspring's phenotype attributable to its own dam. Maternal effects can be classified into ***prenatal*** (e.g. cytoplasm of the egg and uterine environment) and ***postnatal*** (e.g. milk production, method of rearing and/or mothering ability).

Non additive component which we observe as heterosis. It is the amount by which merit in crossbreds **deviates from the additive component**. Heterosis is usually attributed to genetic interactions within loci **(dominance)** and interaction between loci **(epistasis).**

Heterosis or **Hybrid vigour** which is the increase in performance obtained over the mid-parent value when individuals from two different populations are mated together. It is normally expressed as a proportion of the mid-parent value

Population 1	10
Population 2	6
Crossbred	12
Mid-parent value	8
Superiority of crossbred	4
Heterosis	50%

Heterosis is the opposite of inbreeding depression and is caused by dominance effects.

If we cross inbred lines, there is a chance that certain lines will combine in such a way that heterosis occurs in traits of economic importance.

In practise, we don't use inbred lines but cross different breeds or lines bred for specific purposes to utilize hybrid vigour. [contrast this with plant breeding where the use of highly inbred lines is a common technique].

Highly inbred lines are not generally used in animal production because it is mostly ***fitness traits [reproductive traits] that are most affected by inbreeding depression***. In addition, ***there is likely to be a number of unacceptable "mutants" caused by the increase in homozygous recessive genotypes***.

Heterotic effects can be classified into

- Individual heterosis
- Maternal heterosis
- Paternal heterosis

Individual heterosis is the deviation (or superiority) in performance in an individual relative to the average of the parental breeds, with maternal, parental or sex-linked effects playing no role.

Maternal heterosis refers to heterosis in the population attributed to using crossbred instead of purebred dams and occurs due to the dam itself possessing heterosis.
Selection for lines and breeds that show heterosis in maternal traits has been used extensively in pig production.

Paternal heterosis is that due to the sire. It doesn't affect many traits, but is important in traits such as conception rate.

Measuring Hybrid Vigour

$$HV = \overline{P}_{F_1} - \overline{P}_p$$

P_{F1} = The average performance of the crossbreds

P_p = The average performance of both parental lines

$$= \left(\overline{P}_{p_1} + \overline{P}_{p_2} \right) / 2$$

E.g., The average litter weight of purebred pigs

= 98 lbs for breed A

= 100 lbs for breed B

The F_1 average litter weight is 113 lb

Calculate the % hybrid vigour for the cross

Hybrid Vigour can be either positive or negative. A negative HV does not imply unfavourable! It is simply a function of the way the trait is measured.

Eg HV for age at puberty causes animals to reach puberty at an earlier age (hence -ve value)

A crossbreeding system can be defined as a mating system that uses crossbreeding to maintain a desirable level of hybrid vigour and (or) breed complementarity

Finding a good cross

Since heterosis depends on dominance, and dominance effects are the result of the way pairs of particular alleles affect each other at a locus, then as breeders we have to investigate which specific lines or breeds show heterosis.

In addition, we have to consider the best way to use and maintain hybrid vigour in our "commercial" population. [***Crossbreeding systems are the domain of commercial animal production, they are designed to maintain hybrid vigour***

Most seed-stock breeders are pure-bred breeders. They raise stock that is later used in cross-breeding systems, so they need to understand the requirements and needs of the commercial breeders].

In plant production it is possible to generate a large number of inbred lines from a base population and then make all possible crosses between them. The crosses are then evaluated for heterosis. Using such techniques, two components of heterosis are identifiable:

1. The **general combining ability (GCA)**—which is the mean of all crosses derived from a particular inbred line when crossed with all others.

2. The **specific combining ability (SCA)** –which is the mean of the cross between two particular lines over and above the GCA.

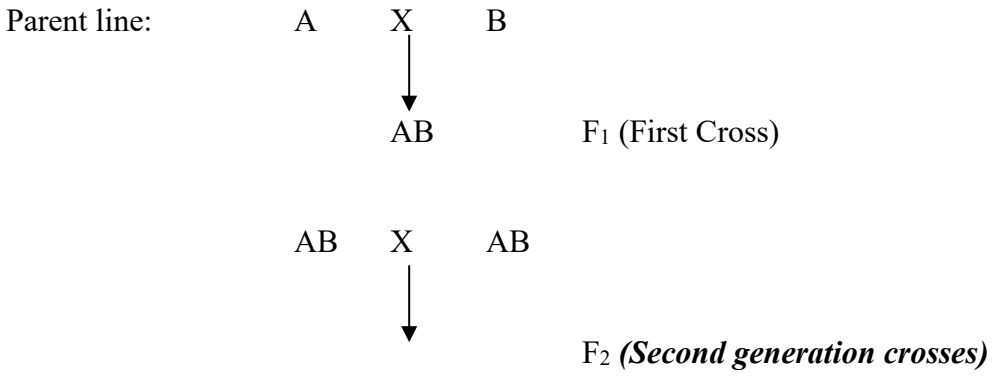
In animals, the testing of a large number of lines is too costly and we tend to develop lines for specific characteristics and then use crossbreeding to combine the different characters developed within the individual lines.

Crossbreeding terminology

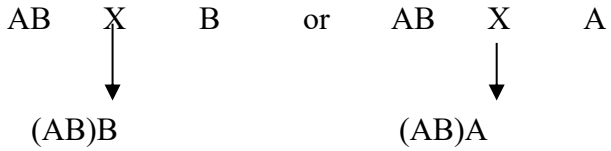
Backcrossing

Assume we have four lines A, B, C, & D.

When we cross A & B, the resulting generation AB is called the first cross and is often abbreviated as F_1:



If we mate an F_1 with one of the parent breeds:



This is referred to as **backcrossing**

Terminal sire crossbreeding

A crossbreeding system in which maternal bred females are mated to paternal bred sires to produce ***progeny that are specifically desirable for the market***. Terminal crossing is most called static crossbreeding in contrast to rotational systems. Terminally sired animals are not kept as replacements but are sold as slaughter animals. Terminal crossing is mainly applied in meat production.

Reciprocal recurrent crossing

A crossbreeding system whereby the animals are ***backcrossed to a different parent line each generation***. This is practised in poultry production.

Systems of breed utilization (mating systems)

Systems of breed utilization are listed in the table below.

Mating system	Symbol
Production with one population	
Pure breeding	
Synthetics: two breeds, balanced	$(AB)_{Syn}$
Synthetics: three breeds, balanced	$(ABC)_{Syn}$
Terminal crossbreeding	
Two-way cross	B*A
Three-way cross	C*BA
Four-way cross	CD*BA
Backcross	B*BA
Rotational crossbreeding	
Two breeds; balanced	$(AB)_{Rot}$
Three breeds; balanced	$(ABC)_{Rot}$
Terminal rotation	$C*(AB)_{Rot}$

Two breeds; breed preference	$(AAB)_{Rot}$
Two breeds; generation preference	$(\underline{A}B)_{Rot}$

Rotational systems

There are many modifications of the rotational crossbreeding systems.

***Terminal rotation*.** When rotational matings are restricted to produce only crossbred dams which are later mated to terminal sires to get a final generation of slaughter animals.

***Rotational system with generation preference*.** Assume that in a criss-cross rotation breed A is more prolific and breed B is a better meat breed. Dams of the more prolific generation (2/3 genes of A) are then kept in a larger proportion than dams from the less productive one. The consequence is that the proportion of slaughter animals of the generation with better meat performance is also higher (2/3 genes of breed B).

***Rotation with breed preference*.** Whenever there are differences in the overall performance of the breeds involved in a rotation, another question arises. Why should the sequence of using sires of each breed be equal? Or would it be beneficial to use the better breed more often to have a higher average gene percentage of it in the system? This would certainly mean an improvement due to breed difference at the expense of a reduction in heterosis. When the criss-cross rotation AB_{Rot} is substituted by AAB_{Rot} (i.e., 2/3 of A genes in the system), the heterosis expressed is 4/7 instead of 2/3. Whenever the overall breed difference is higher than 4/7 of the overall heterosis between the two breeds, this "rotation with breed preference" would be more efficient than the conventional crisscross system.

***Sire rotation without dam identification*.** So far modified systems of rotation have been discussed under the assumption of controlled matings. These assumptions hold usually for pig production and for any commercial use of artificial insemination. In beef cattle herds and sheep flocks, however, there is usually multiple sires joining i.e., in each mating period only sires of one breed should be used. Such a sire rotation, regardless of the genotype of dams they mate, ends up in more heterosis than a synthetic breed with equal gene proportions. The superiority depends mainly on the age structure of the dams. The higher the proportion of younger dams the higher the effect of heterosis through rotation.

***Sire rotation with dam identification*.** There is another alternative procedure of sire rotation under non-controlled matings. Whereas controlled matings might be impracticable in the

commercial sheep and beef industry, a simple dam identification according to its crossbred genotype may be easy to handle. This is obtained by identifying female lamb or calves by notches according to their dams' notch identifying the crossbred generation. In a criss-cross system this may be for example one notch for 1/3 gene proportion of breed A and two for 2/3 gene proportion. Let us again assume that only one sire breed has been used in a mating period, say breed A. Mated to these two genotypes of females, those dams with 1/3 gene proportion of A will give birth to offspring with 2/3 and the other will produce offspring with 5/6 gene percentage of breed A. The first are notched as potential replacements, the second will be considered inferior and determined for slaughter. In the next mating period sires of breed B will be mated which in turn produce potential replacements (1/3 gene percentage of breed A, one notch) and an inferior genotype (1/6 gene proportion of A).

More recently, there has been an increase in the number of hybrid seed-stock, ie F_1's and **composite animals.**

Composites (**or synthetics**) = hybrids designed to retain hybrid vigour without crossbreeding. E.g., Mpwapwa, Kilifi composite, Jamaica hope (produced from crosses with Sahiwal)" animals---ie hybrid animals with at least two or more pure breeds in their background. Composite breeds are expected to be bred to their own kind, retaining a level of hybrid vigour normally associated with traditional crossbreeding systems.

Development of synthetic breeds

The decision to invest in synthetic breed formation in specific situations must be based on consideration of the natural environment (average temperatures, rainfall amount and distribution, solar radiation, relative humidity, soil fertility and the quality of natural forages), the level of management, which is economically feasible, and the relative importance of milk and beef as production goals. Medium and long-term projection of input costs and output values are needed because the development of synthetic breeds has both intermediate and long-term implications (Trail and Gregory, 1981). The steps to be followed are as follows:

Step 1. Decide on the number of breeds and choose those that have complementary effects (breeding values) for the traits of interest to produce the biological type of cattle desired. The choice of the foundation breeds to be used in the formation of synthetic breeds should be based on performance under environmental conditions similar to those in which the synthetic breed will be maintained. Inclusion of the maximum number of breeds of demonstrated potential in the synthetic would result in the retention of the highest proportion of the original heterosis.

Step 2. Decide on the percentage of each breed chosen that you need in the synthetic to obtain the desired biological type. This is determined by the degree of environmental modification. The percentage of heterozygosity retained is high when the contribution of the foundation breeds in the synthetic breed is equal. Hence it is desirable to have equal contributions by each breed if this can be done while at the same time achieving optimum additive genetic composition.

Step 3. Decide a mating scheme to reach the desired percentage of each breed in the shortest period of time. Crossbred bulls are always needed

Step 4. Keep good records (both biological and economic) on the traits of interest for use in calculating expected progeny difference values to guide future selection decisions.

Some examples of synthetic or composite

Composite	Component breeds
Aust. Milking Zebu	Sahiwal, Red Sindhi
Brangus	Brahman, Angus
Murray Grey	Roan Shorthorn, Angus
Santa Gertrudis	Brahman, Shorthorn others
Polwarth	Merino, Lincoln
Warridale	Border Leicester, Merino
Corriedale	Merino, Lincoln
Dorper	Dorset, Persian fat-tail.

Which crossing system to adopt (if any)?

The best crossing system to use depends to a large extent on the additive value of the breeds available as well as the amount of heterosis expressed in crossbred animals. This is illustrated in the table below by describing the conditions under which each crossbred genotype is worthy of choice. The best possible system for maximum genetic merit is always a fully structured system with a short pedigree back to purebreds on both sire and dam sides e.g. F1, backcross, 3 breed cross. However, we usually find that such systems are the most expensive to run.

Mating system	When used
Purebred	When no cross is better
F1 cross	When direct heterosis is important
3 breed crosses	When both direct and maternal heterosis are important
4 breed crosses	When paternal heterosis is important as well
Backcross	When only 2 good paternal breeds are available and/or when direct heterosis is not important
Rotational crosses	When females are too expensive to either buy in or to produce in the same enterprise
Open or closed synthetic	When both males and females are too expensive to buy in. A few initial well-judged importations establish the synthetic, and it can then either by closed (which helps to establish a breed type) or left open to occasional importations.

Evaluation of mating systems

The general model

The performance of crossbreds in a terminal or rotational crossbreeding system can be predicted using estimates of genetic parameters from crossbreeding experiments. Several models have been proposed and used in literature to estimate crossbreeding parameters All these models are based on the general two-locus factorial model of gene effects:

$$\overline{G} = \mu + a + d + aa + ad + dd$$

where $\overline{G}$ is the mean of population *G*, μ describes a general mean, *a* the additive effects, *d* the dominance effect (two-factor interaction within loci), *aa* the additive x additive interactions (two factor interaction between one allele from the first locus and one allele from the second locus), *ad* the additive x dominance interactions (three-factor interaction and includes two alleles from one locus and one allele from the second locus and *dd* the dominance x dominance interactions (four factor interaction between all four alleles in a case of a two-locus system each with two alleles).

The value of crossbreeding

Why cross breeds?

The value of crossbreeding can be split into a number of components:

1. *The averaging of breed effects* - This component is often of negative value where parental breed is superior for the environment of interest. However, in many cases the best type of animal is intermediate in some way, for example (i) an animal of intermediate body size may best be suited to the length of the grazing season, or to market demands and, ii) where two traits are negatively correlated across breeds, and where the, multiplicatively in the expression of economic value. This is the case in many *Bos taurus* (fast growing) x *Bos indicus* (stress resistant) crosses.

2. *Direct heterosis.* Crossbred individuals often exhibit heterosis. Heterosis is measured as the extra performance of the crossbreeds over the weighted average of their parent breeds. The percentage increase in performance ranges about 0 - 10 for growth traits and 5 - 25% for fertility traits. The effect of heterosis on the total production can be even more than this, as effects accumulate over traits.

3. *Maternal heterosis.* Crossbred dams can exhibit considerable heterosis in their ability to raise many, fast growing, viable offspring.

4. *Sire-Dam complementation.* The cost of maintaining breeding females can be very high, especially where fecundity is low. A good crossbreeding system aims to use breeding females, which are small (but not so small for dystocia to be a problem) as well as prolific. When a large breed of sire is used the proportion of feed directed to growing animals is increased, and the production system benefits accordingly.

5. *Possibly cheap source of breeding animals.* This is evident in the Australian prime lamb industry, where surplus Merino ewes are bought cheaply and crossed with Border Leicester ram to produce prime lamb dams. These dams show considerable heterosis for weaning rate, and yet retain reasonable growth characteristics. However, much of the economic value of this crossing system is due to the cheap source of Merino ewes.

6. *Potentially widest use of genetic resources*. The range of genetic material available is greater when more than one breed is involved, and this can give greater flexibility in identifying and developing an efficient breeding programme. Selection intensities can be higher and inbreeding problem can be largely avoided.

Evaluating crossbreeding systems:
There are seven useful criteria for evaluating different crossbreeding systems:

1. Merit of component breeds
2. Hybrid vigour
3. Breed complementarity
4. Consistency of performance
5. Replacement considerations
6. Simplicity
7. Accuracy of genetic prediction.

The relative importance of each criterion varies depending on the production situation.

1. Merit of component breeds

For any crossbreeding system to be effective, the breeds chosen must be considered carefully. Each breed chosen should bring some favourable attributes to the mix.

Eg crossing Holstein Friesian cattle with Boran cattle to be reared within the East African region. The HF have high milk production—whereas the Boran cattle are more adapted to the environmental conditions.

The mean breeding value of each component breed for traits of economic importance should be such that the various breeds complement each other in the system.

Eg Cross HF X Sahiwal

For small holder dairy production, when practising upgrading use: ***Jersey X Zebu***

Hence, one needs to determine the appropriate component breeds, locate available animals of the breeds of interest, purchase the pure-breds, then cross them.

In many cases, seed-stock breeders are not very many, hence the price of pure-breds can be high

2. Hybrid vigour

- Generating hybrid vigour is one of the most important reasons for crossbreeding. The more the hybrid vigour, the better.
- But Maximum hybrid vigour is only obtainable in F_1's --the first cross of unrelated populations
- Most crossbreeding systems do not achieve 100% of the F_1 vigour but maintain acceptable levels by limited back crossing.

3. Breed complementarity

This refers to the production of more desirable offspring by crossing breeds that are genetically different from each other but have attributes that are **complementary.**

It is kind of "mixing and matching"

a) Sire X Dam complementarity--especially important in beef production. Offspring inherit superior market characteristics from their sires, and benefit from the maternal environment provided by their dams

b) Hybrid Seed-stock complementarity-- genetically diverse breeds are crossed to produce a hybrid breeding animal, eg Sahiwal X Ayrshire

4. Consistency in performance

- A crossbreeding system should produce a consistent product. It is easier to market a uniform set of animals than a diverse one (eg in poultry production).
- Also, it is easier to maintain a female population that is basically of one biological type than one made up of several types each with diverse requirements.

5. Replacement considerations

The breeders need to produce a continuous flow of the crossbred animals, hence need pure-bred parents. These can be obtained through

- maintaining pure-bred populations on farm
- purchasing replacements from elsewhere

The decision will depend on the costs involved.

Eg **Rotational crossbreeding systems**—whereby generations of females are “rotated” among sire breeds in such a way that they are mated to sires whose breed composition is different from their own: consider a two-way rotation:

A X B

(AB) X B

F_2 X A

F_3 X B

Rotational systems produce replacement females internally and manage to maintain acceptable levels of hybrid vigour. Several different breed combinations can be used in rotations.

In planning a rotational crossbreeding system, one must ensure that there is consistency in the performance of the crossbreeds, that replacement animals are available for selection, that genetic predictions are accurately carried out for evaluating the various crossbreds, and that the system is simple enough to be practical.

6. Simplicity

- An expensive system that requires an unrealistically high level of management is unlikely to remain in place for long (e.g., Kilifi plantations!!!!)
- Crossbreeding systems should be kept simple and avoid conflicts with other management practices unrelated to breeding

7. Accuracy of genetic prediction

The higher the accuracy of genetic prediction, the lower the selection risk and the more predictable the offspring. Typically, it is sires for which predictions are available, hence using information on accurately evaluated pure-bred sires will make crossbreds attractive to commercial producers.

Genetic prediction

Introduction

The prediction of breeding values constitutes an integral part of most breeding programmes for genetic improvement.

Note:

- Crucial to accurate prediction is the availability of ***records***
- Genetic improvement through selection depends on correctly identifying individuals with the ***highest true breeding value***.

The method employed for the prediction of breeding value (BV) depends on the ***type*** and ***amount*** of information available on the candidates for selection.

Recall (note, terms are used interchangeably depending on which country one is in!):

- ***Estimated Breeding Values (EBV)*** describe the value of an animal's genes to its progeny, yet it transmits only half of its genes to any one progeny. So, the predicted merit of an animal's progeny is half of its EBV. Alternatively, the predicted merit of the progeny of a mating pair is the average of their EBV. Ie. $\boldsymbol{BV_I = ½\ BV_s + ½\ BV_d + m_I}$ (where m_I is the deviation of the breeding value of animal I from the average breeding value of both parents—***Mendelian sampling***)
- ***Estimated Progeny Differences (EPD) = ½ EBV***.

Equations used in predicting BV's and producing abilities always include ***a function of heritability***. This is because heritability indicates how conservative a prediction should be.

The basic prediction equation takes the form:

$$\hat{Y} = \mu_y + b_{y.x}(x_i - \hat{\mu}_x)$$

Where:

Y = the predicted BV (BV_w)

μ = the population mean for the trait (or mean BV in the population = 0)

x_I = (evidence) the first litter performance (P_w)

This equation could be written as:

$$\hat{BV}_{wi} = \mu_{BVw} + b_{BV.P}(P_w - \hat{\mu}_{p_w})$$

Where:

$\boldsymbol{b_{BV.P} = h^2_w = \sigma^2_{BV}/\sigma^2_p}$

Thus $BV_{wi} = 0 + h^2_w(P_w - \mu_{pw})$

If one wants to predict an individual breeding value using evidence other than a single record on the individual, the prediction equation becomes a little more complex depending on the source of information.

Recall: 1. Various methods of selection

Individual or Mass Selection

- Involves selection of individuals on the basis of their own "phenotypic value".
- Individuals to be parents would be those whose performance is closest to the breeding objective
  - This type of selection is good for traits of moderate to high heritability (>0.25)
  - It is also good for traits that are easily measured--eg growth traits.

Pedigree Selection

Pedigree refers to the recording of parents so that the genetic relationship between individuals in a population are known.

In pedigree selection , information on related individuals is used to obtain estimates of breeding values, hence to make decisions on which individuals to select.

Pedigree selection is good particularly for sex limited traits, eg milk yield

Progeny testing

A form of selection in which progeny of individuals are used in estimating breeding values.

Widely used in livestock production--eg in Dairy cattle production, sires are evaluated based on their progeny performance.

- Progeny testing is good for traits that have a low heritability, and for sex limited traits.

Performance testing

- Particularly used in beef production systems.
- Individuals are evaluated based on their performance within a given duration of time.

To carry out a performance test, individuals of similar age are usually brought together to a test station, put on similar conditions, and after an initial period during which they adapt to the new environment, data on specific traits is collected and used to evaluate the animals. Eg Weights collected and growth rate during test period determined, Or, Feed conversion efficiency can be tested.

Selection using information on relatives

Usually, the individual's own performance is not the only source of information about its breeding value. Additional information could be available from its relatives—such as

parents, full sibs, half sibs etc.

In dairy cattle, a bull's breeding value for milk yield can only be calculated based on milk produced by its female relatives—usually its daughters.

Family selection

Whole families are selected or rejected as units, according to the mean phenotypic value of the family. Individual values are thus not acted on except in so far as they determine the family mean.

Consider a population comprising groups of families for example---

We can measure the individuals for a particular trait, and then compare the family mean for each family

2. Correlated characters

Correlated characters are important because:

1. **Pleiotropy** occurs in populations –ie the same genes may affect two different characteristics. (Eg Halothane gene in pigs causes an increased lean yield, but also decreased litter size and lower survival rate.)
2. If we select for a trait that is correlated to a second trait, we need to know what will happen to other trait
3. Under natural selection, correlation of a metric trait with fitness will determine the genetic properties of the character.

Examples of correlated traits include:

Birth weight with weaning weight (beef cattle, sheep and goats)

Milk yield and butter fat % of milk (Dairy cattle)

Feed conversion efficiency (FCE) and growth rate (Broiler chickens, beef cattle)

FCE and back-fat thickness (in pig production)

If two traits x and y are correlated, and we are selecting for x, the correlated response to selection expected in trait y is given by:

$$CR_y = i\, h_x\, h_y\, \Gamma_{Axy}\, \sigma_{Py}$$

Where:

i = the selection intensity for trait x

h_x = the accuracy for trait x ($\sqrt{h^2_x}$)

h_y = the accuracy for trait y ($\sqrt{h^2_y}$)

Γ_{Axy} = the genetic correlation between trait x and y

σ_{Py} = the phenotypic standard deviation for trait y

Methods of Selection for more than one trait

Tandem Selection

Involves selecting for one character for several generations and then changing and selecting for a second character.

If the characters are positively correlated, then progress will be made. If traits are negatively correlated, then one could end up "spinning wheels!"

Independent culling levels

In this instant, one sets minimum values (thresholds) or culling levels for each trait to be selected independently. The individuals selected to be parents are those whose performance is above the culling levels set.

The main disadvantage of this is that an animal is not able to compensate performance in one trait with a positive attribute it has in another, so there may be a lot of "genetic wastage".

The Selection indexes

At its simplest this method involves combining the measurements of two or more characters into a single value for everyone. To obtain the maximum rate of change for all characters under selection simultaneously it is necessary to combine the phenotypic values for each character of an individual in such a way as to account for the relative economic values, the heritabilities, the phenotypic variances and the genetic correlation of the characters.

The Selection index

Hazel (1943) developed a method of combining information from several sources, be it the ***same trait*** from ***different types of relatives*** or ***different traits measured on the animal***, to predict the animal's genetic merit. The method for establishing a selection criterion for a given selection objective is referred to as selection index methodology.

Selection index theory was first developed both as:

- A method for genetic prediction
- A means of combining traits in order to select animals in an economically optimal way.

A) The selection index as a method for genetic prediction

Def: A linear combination of phenotypic information and weighting factors that is used for genetic prediction when performance data come from genetically similar contemporary groups.

A selection index is basically a linear combination of various types of phenotypic information and appropriate weighting factors. It takes the form:

$$I = b_1x_1 + b_2x_2 + \ldots + b_nx_n$$

I = an index value or genetic prediction (EBV, EPD, or MPPA)

b_i = a weighting factor (eg heritability)

x_i = a single item of phenotypic information—eg performance record / the average of a group of performance records

n = the total number of items of phenotypic information

The numerical value obtained for each animal is referred to as an index (I) and is the basis on which animals are ranked for selection.

A genetic index is a measure of an animal's ability to transmit genes on to the next generation. These could be genes for production, conformation or a combination of different traits.

A genetic index is not a measure of a physical trait (eg milk production, stature), but is ***a reflection of its genetic make-up (genotype).*** Many cows could produce high quantities of milk because they have the genetic potential for high milk yield, but the management or environment in which they are reared plays a key role in determining their actual productivity.

A genetic index makes the best estimate possible of an animal's ability to transmit genes to the next generation by discounting the effects of its management regime and other environmental factors. This makes it possible to compare animals in different herds managed under different systems.

If one is interested in the breeding value **for one trait**, the following sources of information can be used:

(a). One or more observations for that specific trait on the animal itself,

(b). Observations for that specific trait on relatives,

(c). Observations for other traits on the animal itself and/or relatives,

Observations from relatives can be useful when

- The number of observations on the animal itself is limited: higher accuracy of the estimated breeding value can be accomplished using information on relatives,
- Traits in the breeding goal can not be observed on the animal itself (e.g. milk production for sires)

Observations on correlated traits are useful when

- they add additional information (increase accuracy),
- they are easier to measure than the trait in the breeding goal.

Selection index is able to combine these different sources of information to estimate the true genetic value.

The relative amount of data from these sources varies—eg unborn animals (pedigree data)

as animals grow then both pedigree data and their own information can be used.

In a selection index, each item of phenotypic information is normally expressed as ***a deviation from a contemporary group mean*** (this ***accounts for environmental differences between contemporary groups***).

The problem with this approach is that it ***assumes all contemporary groups are genetically similar***. If the mean breeding value of contemporary groups differ, then the use of contemporary group (CG) means creates bias in the data.

[def: Bias = any factor that causes distortion of genetic predictions]

Bias because records from genetically poorer CG's appear better than they should, and records from genetically superior CG's appear worse than they should.

The selection index is affected by :

The heritability of the trait
The family size
Phenotypic variation
The relationship between members of the population

If several assumptions are fulfilled, selection index will be an optimum procedure. These assumptions include:

- linearity of economic values,
- normally distributed observations and genetic values,
- conditional means (i.e. fixed effects such as herd or season) known without error (unbiasedness).

Given these assumptions, the **selection index is the best estimator of genetic value,** because it fulfils the following criteria

- it maximizes the correlation between true genetic value for the aggregate genotype (H) and the estimated genetic value (index value I),
- it maximizes the probability of a correct ranking of the animals according to their true breeding value
- it maximizes genetic gain by selection,
- it minimizes the differences between true breeding value and estimated breeding value.

A selection index is the best linear prediction (BLP) of an animals breeding value, and takes the form of the multiple regression of breeding value on all sources of information.

For example—if we have a phenotypic value P for an individual as a deviation from the population mean

The breeding value (BV) $= h^2 P$

Since h^2 can also be defined and the regression of the breeding value on the phenotypic value,

$BV = b_{bv} P$

If we have two pieces of information on the individual, then the expected breeding value is:

$BV_I = b_1P_1 + b_2P_2$

The regression coefficients change based on the amount and source of information available-eg :

Source of Information	**Regression coefficient (*b*)**
Single performance record on individual	h^2
The average of **n** records on the individual	$nh^2/ 1+(n - 1)r$
The average of single records on **m** half-sibs	$mh^2 / 4 + (m - 1) h^2$

The average of single records on **p** progeny	$2ph^2 / 4 + (p - 1) h^2$

***(Refer to table 11.1 given for more detailed information on regression coefficients)

Regressing of genetic prediction for amount of information results in **ubiased predictions**.
By regressing predictions for amount on information, predictions are in essence adjusted for accuracy—ie the accuracy of a prediction is accounted for in the prediction itself.
This has the advantage that it allows direct comparison of prediction for different animals regardless of the accuracy of those predictions.
Eg if two animals have the same EBV, but the accuracy of on EBV is higher than the other, one is still justified in considering both as genetic equals.

Example:

3 Dairy cows:

Cow name	No. of indiv lactation records	Avg of all Lact above Contemporary group mean
Iris	5	1072lb
Rose	1	-918lb
Violet	2	1204lb

Assume heritability = 0.25, repeatability =0.5
Calculate the MPPA for milk yield for each cow, and the accuracy with which each index is estimated.
Solution:

Note:

- Accuracy is different depending on the amount of information available to obtain the estimate
- The MPPA is lower than the actual ($x - \mu$). In essence the productivity of each cow has been regressed for the amount of information used/ available.

Accounting for common environment

Common environmental effects are important especially when dealing with full-sib families (eg in pig production). Common environmental effects also occur in paternal half-sib families of dairy sires (a farmer may treat daughters of one sire differently from daughters of another sire) If not accounted for, common environmental effects ***may bias*** genetic predictions.

To account for this effect, a measure of ***covariation among relatives (c^2)*** is incorporated into the analysis. (eg row 6 and 9 of table 11.1)

Assignment

A boar has sired 20 litters averaging 7.8 weaned pigs each. Post-weaning average daily gain of his progeny average 0.06lb per day above contemporary group means. Heritability for post-weaning gain is 0.28, and C^2_{fs}, a measure of covariation among post-weaning gains of littermates that is caused by common environment (covariation due to a common dam) is estimated to be about 0.07.

a) Use a single source selection index to estimate the boar's breeding value for post-weaning ADG
b) Estimate the boar's breeding value assuming there is no environmental covariation among post-weaning gains of littermates
c) Why did his EBV increase when C^2_{fs} was assumed to be zero?

Note:

- Selection indexes should only be used for genetic prediction when performance data come from CG's thought to be genetically similar (eg used in poultry production)
- Though restricted in application, selection indexes are useful for within herd/ flock evaluations.
- Factors affecting accuracy of predictions include:
  Heritability
  Pedigree relationship
  Number of records available

Recall:

Selection index is able to combine different sources of information to estimate the true genetic value. The selection index can be presented as follows:

$$I = b_1X_1 + b_2X_2 + \ldots\ldots\ldots\ldots + b_nX_n$$

where,

I = index value - the estimation of the genetic value for the aggregate genotype,

$X_i = i^{th}$ source of information - one observation or an average of several observations, expressed as a deviation from a conditional mean,

b_i = weighting factor for the ith source of information

Essential point in estimating breeding values, using the selection index, is the derivation of weighting factors (b-values) for the information sources, that fulfil the above mentioned criteria. Next to that, one can derive the accuracy of the estimated breeding value and predict the relative change in the different traits in the breeding goal (selection objective). Selection index theory is very flexible and offers great opportunities in predicting accuracy of estimating breeding values. Consequently it plays an important role in prediction of genetic response of breeding schemes and the design of breeding schemes.

B) Selection Index: As a means of combining traits in order to select animals in an economically optimal way.

Animal breeding can be defined as the genetic manipulation of biological differences between animals over time using approaches aimed at maximizing profitability.

Economic questions that could arise include:

- What is the direction of change which maximizes profitability?
- What is the combination of differences, for traits, between animals which maximizes returns over costs?
- How is this direction established?

To attempt to answer these questions, Economic Selection Indexes are used.

Definition: An economic selection index is combination of weighting factors and genetic information (either phenotypic data or genetic predictions) on more than one trait. Economic selection indexes are used in multiple trait selection to predict aggregate breeding value.

The index looks the same as before:

$$I = b_1X_1 + b_2X_2 + \ldots\ldots\ldots\ldots + b_nX_n$$

where,

I = index value - the estimation of the genetic value for the aggregate genotype,

X_i = i^{th} source of information - one observation or an average of several observations, expressed as a deviation from a conditional mean,

b_i = weighting factor for the ith source of information

However, there are two essential differences between the economic selection index and the index for predicting breeding values:

- The index value in this case is no longer a genetic prediction for a single trait, but rather, it is a prediction of the aggregate breeding value
- The "evidence" in this case is often a genetic prediction (EBV or EPD) and not an individual item of phenotypic information

The weighted combination of traits defining aggregate breeding value is called the **Breeding Objective.**

(Definition: Breeding Objective is a weighted combination of traits defining aggregate breeding value for use in an economic selection index)

The breeding objective sets the breeding direction and provides the basis against which the effectiveness of a breeding program can be gauged.

Assume that there are several traits which have to be improved, denoted by Y_1, Y_2, …. Y_n and the traits have ***economic values*** of a_1, a_2, ..., a_n, respectively.

Definition: The ***economic value of a trait represents the additional economic return per marginal unit improvement in the trait***.

For example, the economic value of carcass lean content may be two economic units, assuming no change in food intake or growth rate.

Since the **economic objective** is to improve all traits, then the traits and their economic values are combined into a **breeding objective** (or selection objective), where

Breeding objective (H) $= a_1Y_1 + a_2Y_2 + \ldots\ldots\ldots\ldots + a_nY_n$

The breeding objective can be expressed in matrix notation as a'Y and is denoted by H.

H = aggregate breeding value

a_i = an economic weight for a trait in the breeding objective

Y = the breeding value for a trait in the breeding objective

n= the total number of traits in the breeding objective

For example, if growth rate and carcass lean content are to be improved and have economic values of 5 and 2, respectively, then the breeding objective is

5 x growth rate + 2 x carcass lean content.

Note:

The traits that appear in the breeding objective should be those that are economically important.

A distinction needs to be made between traits of the breeding objective, which are desired to be improved, and the available measures or **Selection Criteria**, which are useful because of the information they provide about the traits.

Selection criterion

Traits for which information can be easily and cheaply collected which either contribute or are related to the breeding objective are included in the selection criteria.

The traits which are measured to predict the breeding value are denoted X_1. X_2, ..., X_m.

The measured traits are combined into an **index** on which the animals are selected. The selection index or selection criterion is follows:

selection criterion (I) = $b_1X_1 + b_2X_2 + \ldots\ldots\ldots\ldots + b_nX_n$

The selection criterion can be expressed in matrix. notation as b'X and is denoted b I (for index).

For example, if growth rate and ultrasonic backfat depth are the traits measured, then animals are selected on the basis of

b_1 x growth rate + b_2 x ultrasonic backfat depth.

Note:

- Traits in the ***selection criterion need not necessarily be the same traits in the breeding objective***. Similarly, the number of traits in the ***breeding objective*** need not be the same as the number of traits in the selection criterion. ***For example***, in one breeding programme, growth rate and carcass lean content are the traits to be improved, while growth rate and ultrasonic backfat depth are the traits measured for selection purposes. In a second breeding programme, litter weight at weaning is to be improved (breeding objective), and the selection criterion consists of litter size and weight at birth.
- ***The traits in the breeding objective should be decided by their importance irrespective of their ease or cost of measurement***. Economically important but otherwise troublesome to measure traits in the breeding objective are often replaced with indicator traits. For example, a breeding objective for beef cattle might include: yearling weight, age at puberty,

and calf death loss at birth. The corresponding selection criteria might include, yearling weight, scrotal circumference, and birth weight (scrotal circumference serves as a proxy for age at puberty, and birth weight serves as a proxy for calving losses)

The selection index method, determines the selection criterion coefficients that maximize the response in the breeding objective, H, with selection on the selection criterion, I.

As several traits can be included in both the selection objective and the selection criterion, then information on the ***variance of the traits*** and ***on relationship between the traits*** at the phenotypic and genetic levels is required.

Calculating the Classic form of Economic selection Index

There are several kinds of economic selection indexes.:

1. Phenotypic selection index

These are used in situations where selection is strictly based on the individual's own performance (no relatives)

In the classic form of a phenotypic index, the traits in the selection criteria are the same as those in the breeding objective.

For example, a breeding objective involving yearling weight (YW)

H = Xyw + Zbw

I = yw + bw

The selection index method determines the selection criterion coefficients that maximize the response in the breeding objective (H) with selection on the criterion (I)

Several traits can be included in both I and H.

What is required is information on

- ***Variances of the traits***
- ***Relationships between the traits, both phenotypic and genetic***

The information is in the form of three matrices:

P: the phenotypic variance-covariance matrix of traits in the selection criterion.

G: the genetic covariance matrix between traits in the breeding objective and the traits in the selection criterion

C: the genetic variance-covariance matrix of traits in the breeding objective.

S= Selection criteria trait = two traits S1 & S2

BJ = Breeding Objective trait = two traits BJ1 & BJ2

P =

	S1	S2
S1	VAR_P (S1)	COV_P (S1, S2)
S2	COV_P (S2, S1)	VAR_P (S2)

=

	S1	S2
S1	VAR_P (S1)	r_P x σ_P S1 x σ_P S2
S2	r_P x σ_P S1 x σ_P S2 $\{\Gamma_{xy}\sigma_x\sigma_y\}$- *see below*	VAR_P (S2)

$h^2 = V_A / V_P$

$$V_p * h^2 = \frac{V_a}{V_p} * V_P$$

$V_p * h^2 = V_a$ $= V_P * h^2$

$$V_p = \frac{V_a}{h^2}$$

$V_P * h^2 = V_A$ .....OR.....

$\sigma_A{}^2 = h^2\sigma_P{}^2$

SQRT ($V_P * h^2$) = SQRT (V_A)

$\sigma_A = \sigma_p * h$

$\sigma_p = \sigma_A / h$

G =

	S1	S2
BJ1	COV_A (BJ1, S1)	COV_A (BJ1, S2)
BJ2	COV_A (BJ2, S1)	COV_A (BJ2, S2)

Example where S2 is both the breeding objective and selection criteria trait)

	S1	S2*
BJ1	COV_A (BJ1, S1)	COV_A (BJ1, S2)
S2*	COV_A (BJ2, S1)	VAR_A (S2)

* *S2 is both a breeding objective and selection criteria trait*

=

	S1	S2
BJ1	r_A x h $_{BJ1}$ x h_{S1} x $\sigma_{P\ BJ1}$ x $\sigma_{P\ S1}$	r_A x h $_{BJ1}$ x h_{S2} x $\sigma_{P\ BJ1}$ x $\sigma_{P\ S2}$
BJ2	r_A x h $_{BJ2}$ x h_{S1} x $\sigma_{P\ BJ2}$ x $\sigma_{P\ S1}$ $\{\sigma_A = \sigma_p * h\}$	r_A x h $_{BJ2}$ x h_{S2} x $\sigma_{P\ BJ2}$ x $\sigma_{P\ S2}$

C=

	BJ1	BJ2
BJ1	VAR_A (BJ1)	COV_A (BJ1, BJ2)
BJ2	COV_A (BJ2, BJ2)	VAR_A (BJ2)

=

	BJ1	BJ2
BJ1	h^2_{BJ1} X $\sigma_P{}^2_{BJ1}$ $\{\sigma_A{}^2 = h^2\sigma_P{}^2\}$	r_A x h $_{BJ1}$ x h_{BJ2} x $\sigma_{P\ BJ1}$ x $\sigma_{P\ BJ2}$ $\{\sigma_A = \sigma_p * h\}$
BJ2	r_A x h $_{BJ2}$ x h_{BJ1} x $\sigma_{P\ BJ2}$ x $\sigma_{P\ BJ1}$	h^2_{BJ2} X $\sigma_P{}^2_{BJ2}$

where var_P and cov_P are phenotypic variance and covariance, var_A and cov_A are additive genetic variances and covariances, r_P and r_A are the phenotypic and genetic correlation between traits and is the phenotypic variance for a trait.

Recall:

$h^2 = b_{BV.P}$

$h^2 = \sigma_{BV}{}^2/\sigma_P{}^2$

thus: $\sigma_{BV}{}^2 = h^2\sigma_P{}^2$

Correlation between two traits

$$\Gamma_{xy} = \mathbf{Cov\ (xy)/\sigma_x\sigma_y}$$

$$Cov\ (x,y) = \Sigma(x-\mu_x)\ (y - \mu_y)/n = \Gamma_{xy}\sigma_x\sigma_y$$

$$b_{y.x} = Cov\ (x,y)/\sigma^2_x$$

The P and C matrices are always symmetric. The G matrix will not generally be a symmetric matrix, as the number of traits measured will not always equal the number of traits to be improved and, secondly, the traits in the selection objective will not necessarily be the same traits as in the selection criterion.

G-matrix, C-matrix

Elements of the G-matrix and the C-matrix are easier to compute than those of the P-matrix; only genetic variances are involved. In general, G-matrix elements are as follows

$R_{ij} * r_{Aij} * \sigma_{Ai} * \sigma_{Aj}$

where R_{ij} = the additive genetic relationship between the animal used for observation and the animal, for which a breeding value is estimated. Elements of the C-matrix are given by

$r_{Aij} * \sigma_{Ai} * \sigma_{Aj}$

σ_{Ai} represents the genetic standard deviation for the ith trait.

The classic selection index:

$$H = a_1A_1 + a_2A_2 + \ldots\ldots\ldots\ldots + a_nA_n$$

$$I = b_1X_1 + b_2X_2 + \ldots\ldots\ldots\ldots + b_nX_n$$

Properties of I

- It minimizes the average square prediction error
- It maximizes the correlation between true BV and the index

To obtain b values, a set of simultaneous equations are obtained and solved for b.

Ie

$b_1P_{11} + b_2P_{12} + \ldots + b_mP_{1m} = g_{11}$

$b_1P_{21} + b_2P_{22} + \ldots + b_mP_{2m} = g_{12}$

.

.

.

$b_1P_{n1} + b_2P_{n2} + \ldots + b_nP_{nn} = g_{1n}$

$P_{ii} = \sigma_P^2$

$P_{ij} = Cov_p\ (a_I\ ,a_j)$

$g_{ii} = \sigma_{ai}^{\ 2}$

$g_{ij} = Cov_g\ (a_I\ ,a_j)$

In matrix form:

Pb = G

P= variance covariance matrix of observations

G = covariance matrix between observations and BV's to be predicted

Thus:

$\mathbf{b = P^{-1}G}$

$\mathbf{I = P^{-1}G}(y - \mu)$

$= \mathbf{b}'(y - \mu)$

To get b values, one needs to set up P and G matrices

Example

Assume the following parameters were obtained for average daily gain (ADG) from birth to 400 days, and lean percentage (LP) at the same age group for beef calves:

Trait	heritability	Standard deviation
ADG (g day^{-1})	0.43	80.0
LP %	0.30	7.2

If the ***genetic*** and ***phenotypic*** correlation between ADG and LP are ***0.3 (r_A)*** and ***-0.10 (r_P)*** respectively, construct an index to improve growth rate in the beef calves.

Breeding objective trait = ADG

Selection criteria = ADG & LP

Solution:

Assume ADG = Trait 1

LP = Trait 2

Thus,

$I = b_1X_1 + b_2X_2$

Equations to be solved:

$b_1P_{11} + b_2P_{12} + \ldots + b_mP_{1m} = g_{11}$

$b_1P_{21} + b_2P_{22} + \ldots + b_mP_{2m} = g_{12}$

In Matrix form:

$P_{11} = \sigma_P^2 = 80^2 = 6400$

$P_{22} = \sigma_P^2 = 7.2^2 = 51.84$

$P_{12} = Cov_p\ (a_I\ ,a_j) = \Gamma_p\sigma_{P1}\sigma_{p2} = -0.1 \times 80 \times 7.2 = -57.6$

P - MATRIX

	ADG	LP
ADG	VAR_P (ADG)	COV_P (ADG, LP)
LP	COV_P (ADG, LP)	VAR_P (LP)

=

	ADG	LP
ADG	VAR_P (ADG)	r_P x σ_P ADG x σ_P LP
LP	r_P x σ_P ADG x σ_P LP	VAR_P (LP)

=

	ADG	LP
ADG	$\sigma_P^2 = 80^2 = 6400$	$\Gamma_p\sigma_{P1}\sigma_{p2} = -0.1 \times 80 \times 7.2 = -57.6$
LP	$\Gamma_p\sigma_{P1}\sigma_{p2} = -0.1 \times 80 \times 7.2 = -57.6$	$\sigma_P^2 = 7.2^2 = 51.84$

G- MATRIX

$g_{ii} = \sigma_{ai}^{\ 2} = h^2\sigma_P^2$

$g_{11} = 0.43 \times 6400 = 2752$

$g_{22} = 0.30 \times 51.84 = 15.552$

$g_{ij} = Cov_g\ (a_I\ ,a_j) = \Gamma_g\sigma_{ai}\sigma_{a2}$

$g_{12} = 0.3 \times \sqrt{(15.552 \times 2752)} = 62.064$

	BJ1	S1
BJ1	h^2_{BJ1} X $\sigma_P{}^2_{BJ1}$	r_A x h_{BJ1} x h_{S1} x $\sigma_{P\ BJ1}$ x $\sigma_{P\ S1}$

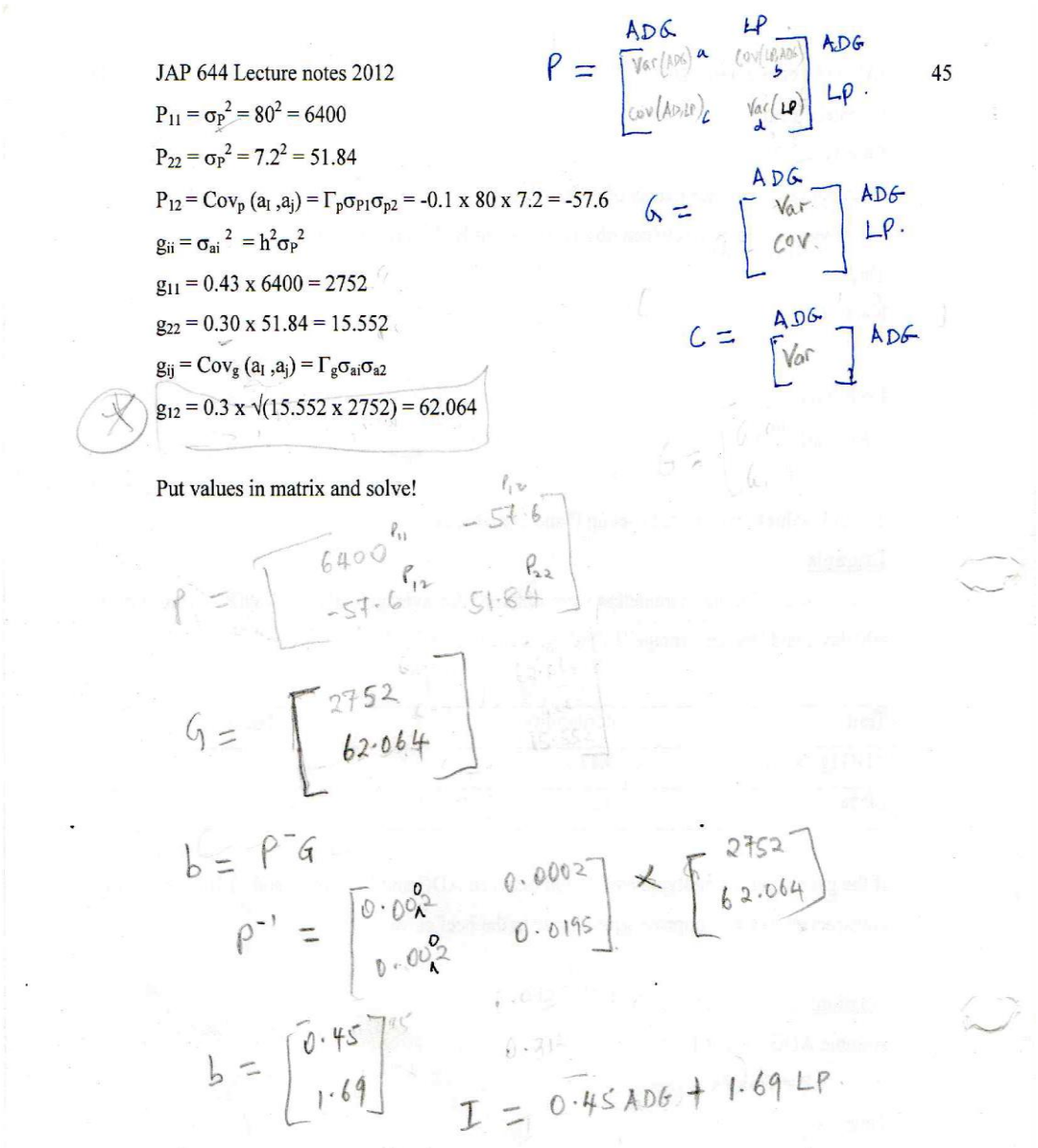
=

	ADG	LP
ADG	h^2_{ADG} X $\sigma_P{}^2_{ADG}$	r_A x h $_{ADG}$ x h_{LP} x $\sigma_{P\ ADG}$ x $\sigma_{P\ LP}$

=

	ADG	LP
ADG	0.43 x 80^2 = 2752	r_A x h $_{ADG}$ x h_{LP} x $\sigma_{P\ ADG}$ x $\sigma_{P\ LP}$ 0.3 *√0.43 *√0.3*√80^2 *√7.2^2 0.3*0.656*0.548*80*7.2

Put values in matrix and solve!



Accuracy of Index (Γ_{aI})

The accuracy of an index is the correlation between the true breeding value and the index. The higher the correlation, the better the index as a predictor of breeding value.

Accuracy is important because it provides a means of evaluating different indices, based on different observations-- to find out if for instance, a particular observation is worth including in an index or not.

Index II: Using single records on individual and relatives.

Suppose the ADG for a bull calf (y_1) = 900 g day^{-1}, and the ADG for his Sire (y_2) and dam (y_3) are 800 g day^{-1} and 450 g day^{-1} respectively. Assuming that all observations are obtained in the same herd, predict the breeding value for the bull calf for ADG and its accuracy

	Heritability	Standard deviation
ADG g Day^{-1}	0.43	80.0

Solution:

Predicting the aggregate genotype (H)

Often various traits are measured in animals. If selection is for one trait, the information of all traits measured can be used to estimate breeding value for the trait under selection. If the other traits have a non-zero genetic correlation with the trait under selection, these traits provide additional information. In a multiple trait breeding goal, the breeding goal is a linear combination of genotypes of several traits. As alluded to, the breeding goal (H) is formalized by an aggregate genotype, which is written as

$$H = a_1A_1 + a_2A_2 + \ldots\ldots\ldots\ldots + a_nA_n$$

Where A_i is the breeding value for trait i and a_i is a relative weight. The weight a_i indicate the relative value of one unit of genetic change of a certain trait. In most applications where breeding is an economic undertaking, the relative weights for a_i represent economic values.

The economic value represents the value of improving a trait by one unit on all animals in the population. The value does not only account for the gain in efficiency by improving the trait in one animal, but also for the frequency and the time at which the trait is expressed. The frequency of expression depends on the number of animals that express the trait (e.g. only females, only reproductive animals) and the number of times a trait is expressed by one animal (e.g. lactation: 1st , 2nd , 3rd .... lactations, carcass quality). The time of expression accounts for the lower value of returns to investment if they are obtained later in a time horizon. If breeding is considered as an investment activity, a comparison should be made between traits that can be improved on the short term vs. improvement of traits where the

profit only arises after a long period. However, often, traits are expressed more or less at the same time and in that case no account for time horizon is needed.

Deriving economic values is basically the most important activity in defining the breeding goal. This activity does not always get the attention it deserves. Formally, the economic value is the first derivative of a profit function to the traits to be improved. An important assumption in the selection index theory is that this profit function is linear, and therefore, that the economic value of a trait does not depend on the mean level in the population.

Although in principle there could be many traits that could have an economic value different from zero, the breeding goal contains usually a limited number of traits. For traits to be included in the breeding goal, phenotypic, genetic and economic parameters have to be known. Such parameters are usually estimated only for easily measurable traits like production and reproduction traits. For other traits e.g. disease resistance underlying physiological traits etc. usually accurate parameter estimates are not known and it is difficult and unpractical to include

such traits in the breeding goal. It should be considered, however, to regularly monitor if changes in these traits have occurred. Another argument not to include too many traits is that the additional gain from including more traits may be small.

$$\mathbf{H = w_1a_1 + w_2a_2 + \ldots + w_ma_m}$$

a_i = EBV for the ith trait

w_i = weighting factor

The principles are the same as for I, but in this case the relative economic weight for each trait is included.

Thus in matrix form:

$$\mathbf{H = P^{-1}Gw\ (y - \mu)}$$

W = vector of economic weights.

The equations to be solved in this case are:

$b_1P_{11} + b_2P_{12} + \ldots + b_mP_{1m} = w_1g_{11} + w_2g_{12} + \ldots + w_mg_{1m}$

$b_1P_{21} + b_2P_{22} + \ldots + b_mP_{2m} = w_1g_{11} + w_2g_{12} + \ldots + w_mg_{1m}$

etc

or

$$\mathbf{Pb = Gw\ ,\ thus,\ b = P^{-1}Gw}$$

Note:

1. Because traits in the index are not always the same as traits in the breeding objective, the vector of economic weights may not necessarily be of the same dimension as traits in the index.
2. Each trait in the index is weighted by the economic weight relevant to the breeding value of the trait it is predicting in the aggregate breeding value (H)

The equation:

$$\mathbf{H = P^{-1}Gw\ (y - \mu)}$$

Implies that a change in economic weight for one trait would necessitate recalculating the index!

This equation can be re-written:

$$\mathbf{H = \Sigma\ I_iw_I}$$

Where $\mathbf{I_I} = \mathbf{P^{-1}G(y - \mu)}$

And $\mathbf{w_I}$ = economic weight

In this case, a change in the weight can easily be implemented without recalculating the index.

Example:

Assume the economic weight for ADG g day^{-1} and LP% are £1.50 and £0.50 for an increase in 1 kg in ADG and a 1% increase in LP respectively.

Using genetic parameters in previous example, construct an index to select for fast growing lean beef calves using:

a) $\mathbf{H = P^{-1}Gw\ (y - \mu)}$

b) $\mathbf{H = \Sigma\ I_i w_I}$

Solution

Selection index method is easier to work out if the following 5 steps are taken consecutively;

1. **Definition of index and breeding goal**
2. **Setting up of variance-covariance matrices**
3. **Calculation of index weights**
4. **Variance of index and breeding goal.**
5. **Calculation of accuracy**

Steps in defining the breeding objective

- Identification of the breeding, production and marketing system

This involves such things as whether animals are to be purebred or cross-bred, the characteristics of the production unit and the final product; identifying any prevailing restrictions on the production or marketing.

- Identification of sources of returns (R) and costs (C) in the herd

  This allows developing a function describing the profit, or economic efficiency of production:

  i.e P = R - C
- Determination of the biological traits influencing returns and costs
- Derivation of the economic value of each trait

  The change in profit resulting from a unit change in each trait is determined assuming all other traits remain constant.

Example:

Dairy Cattle.

Traits of economic importance (see list for some examples)

It is a little presumptuous to give a single objective/ goal for all dairy producers.

Managers differ in:

- Skills required for dairy business
- Preferences
- Short and long term targets

What is common among all is:

- they are business people
- They are all entrepreneurs searching for economic gain or economic survival

Thus its important for every manager to know:

a) How much is earned
   - in animal sales (cull cows, breeding stock
   - in milk sales
b) How much cost is incurred in each category of operations
   - Feed
   - Labour
   - Equipment
   - Supplies
   - Semen

Although the main objective may be maximum profit, sometimes decisions are made to ensure economic survival in bad years by minimizing losses!

Overall, the challenge is generally to maximize the total annual net economic returns. One must thus think ***of investment strategies***--eg genetically improve cows by selecting AI sires in a way that will yield high or maximum net economic returns.
Financial planning requires predicting various ***budgets***, hence managers need to apply sound methods to guide their decisions rapidly and accurately.

In the breeding Objective:

- Information ot select sires may be classified as the traits affecting
  - income from milk sales
  - costs of milk production
  - sale value of breeding stock

PTA for traits related to milk income from future daughters are:

- composition
- yield of milk
- somatic cell count

- Genetic information about the incidence of calving difficulty (dystocia) in current matings will also affect the cost of producing milk, hence

PTA for type traits and conformation score

- Managers wanting profit from their genetic decisions are those who expect more net income from genetic gains form selected Sires than it will cost them to buy semen to produce their daughters and descendants

Note: The increase in net earnings form a daughter (or descendant) of a selected sire is called the ***marginal value of product***
The increase in cost of semen to obtain the genetic gain is called the ***marginal factor cost***

The economic principle or rule to maximize net income with a limited budget is to choose between investment alternatives (in this case AI sires) with the greatest difference between marginal value of product and marginal factor cost.

There is a ***time lag*** between the genetic improvement **decision** to purchase semen from a selected sire and the **income earned** in the various lactations of their daughters. Consequently,

the stream of costs and returns needs to be standardized by using appropriate multiplicative discounting factors to put them on a comparative time basis.

The theoretically correct method for equating marginal value of product and marginal factor cost that always gives the most profitable investment decision is to maximize ***Net Present Value (NPV)*** of the costs and the returns. The ***discounting factor*** increases with time at a decreasing rate (geometrically)!

After considering Semen price, to correctly specify a breeding objective requires:

Accurately determining factors influencing economic returns:

Ie.

- Accurately specifying conception rates for different age classes of animals
- Determining a financial planning horizon for each age class
- Determining the number of semen units required to produce female replacements
- Information on sires about calving difficulties (this is important as it costs to loose a calf, it costs to have a vet supervise calvings!)

Accurately determine how the factors influencing economic returns are likely to change for the unique circumstances of each farm

Accurately assess incomes associated with various products.

In developed countries, there are available various selection indexes that producers can choose to adopt to assist them make the decisions required. Eg

- Indexes aimed a increasing genetic response
- Indexes aimed at genetic increase after imposing relevant restrictions
- Indexes that maximize the net economic returns from genetic improvement.

Complexities of economic selection indexes

- Determination of economic weights

  Economic weights require careful analysis of costs and returns and are likely to be different in different situations.
- The indices depend on the level of production, and this is tricky to determine especially in traits for which an optimal level of production is intermediate--eg in dual purpose animals
- The indexes change over time:
  - costs and prices change with fluctuation in markets
  - management changes with the introduction of new technology

- levels of production change with genetic trend

  hence weights need to be adjusted/ updated from time to time

- Selection indexes provide direction for selection, but give no indication of a selection end point for the individual traits in the index--ie they don't provide "selection targets", hence their use can occasionally result in individuals that are dangerously extreme in one or more traits.

(Targets provide breeders with concrete goals, while economic indexes offer the most efficient way to achieve these goals!)

Ultimately, selection should be conducted with the end user in mind. Hence, selection targets, culling levels and economic weights should reflect the circumstances of the commercial producers.

Response to multiple trait selection

The total response to index selection is equal to

$R = i \times r_{IH} \times \sigma_I$

The response is therefore determined by the accuracy of the index and the standard deviation of the breeding goal. This total response is aggregated over all breeding goal traits, and usually expressed in monetary terms.

Note:

The more traits selected for in the selection index, the lower the selection intensity and the lower the rate of genetic change in any one trait. Thus the number of traits in an index should be kept to a minimum.

When choosing a trait for a breeding objective, questions of interest are:

- By what proportion will the trait contribute to the aggregate breeding value?
- Does the trait contribute to long / short term economic benefits? (traits with short term economic benefits in the long run reduce progress in other more important traits)
- Are economic benefits of including the trait worth the cost of recording?

DESIGN OF BREEDING PROGRAMMES

Introduction

Animal breeders aim to bring about genetic change in their livestock, with a view to:

- Increase profitability
- Ease management
- Sustain production

This involves a large number of issues, which can be classified into three groups:

- **Objectives to target:** This relates to asking 'Where to go?.

  Breeding objectives describe the type of animal that one might want to develop, or, more commonly, describe the utility of marginal genetic changes in each trait of commercial value.

  However, there is increasing concern about ***loss of genetic diversity***, both within and between breeds, and this also needs to be accommodated in our objectives and breeding methods.

- **Tools to exploit:** Tools and Strategies relate to 'How to get there?

  The animal breeder's toolkit includes:

- Performance and pedigree recording systems,
- Novel reproductive techniques,
- Knowledge of population genetic parameters,
- Genetic evaluation systems.
- Emerging tools include information from genetic markers

- **Strategies to adopt:** This relates to the ***design of breeding programmes.***

  In terms of implementing breeding programmes, the main components are deciding:

- Which animal to use as parents
- In what combination should they be mated

  These are the driving elements of both ***selection theory*** and ***crossbreeding theory***.

  Other strategy decisions involve:

- Data recording (pedigree, trait performance and DNA test information),
- Use of novel reproductive and gene transfer techniques.

Breeding program design dictates the animal selection and mate allocation decisions made by animal breeders.

In many cases, design also dictates ***the biology of mating*** (eg. use of AI and MOET), ***location of matings*** (as affected by migration of seedstock between farms) ***timing of mating***, and the ***pattern of trait recording***.

What is the best design to use?

There are various criteria or issues by which we can judge how good any one design is. Some of these are outlined in the following table. There is also a very wide range of breeding strategies and designs to choose from, and it is often difficult to predict their effect on each of the issues in the table below.

Issue	Comment
Genetic gain	This is increase in genetic merit over time, either for a given trait, or for a group of traits in a breeding objective. Genetic gain is usually driven by selection on Estimate Breeding Value.
Inbreeding	Inbreeding is caused by the mating of relatives, and it results in loss of merit through inbreeding depression, and loss of the genetic variation that fuels genetic gains. Breeding schemes that attempt to exploit inbreeding are generally ineffective.
Crossbreeding value	This is the merit of crossbred animals, which can be predicted from direct and maternal breed differences and heterosis.
Connection	Different herds (or other groupings) can be connected by reference sires or other genetic links. This gives power to genetically compare animals from different groups, resulting in better genetic gain in later generations.
Assortative mating	This is mating best to best within the selected groups of animals. Such elite matings, as used in open nucleus schemes, give improved long-term genetic gains, but at some risk of higher inbreeding.

Measurement strategies	This includes trait measurement and even DNA testing. It is best to invest most measurement where it is most effective. For example, males are often more important in this regard as they can be selected on more intensely. Multi-stage selection can be used to reduce measurement costs.
Quantitative Trait Loci	Both detection and use of 'major genes' can be aided considerably by appropriate program design.
Lifetime value	Many breeding operations gain much of their income from sale of product. The best cow to use in the breeding program is not necessarily the best cow for lifetime production in the herd
Reproductive manipulation	It has been said that multiple ovulation turns a cow into a sow. This shifts the ideal design from one appropriate for cattle to one more appropriate for pigs.
Running costs	The cost of running a breeding program is all too often ignored when it is being designed. Costs include feeding costs (reflected in herd size), purchase of seedstock, sernen and embryos, and transport costs.
Risk	For many breeders, staying in business is as important as making large profits. This often leads to conservative breeding program design, based on more parents, with less variance in outcome at the expense of reduced probability of very large gains
Breeding objectives	This is probably the most important component of all. The breeding objective determines the direction of genetic change. A powerful breeding design is of little use if it goes in the wrong direction.

Nucleus breeding schemes

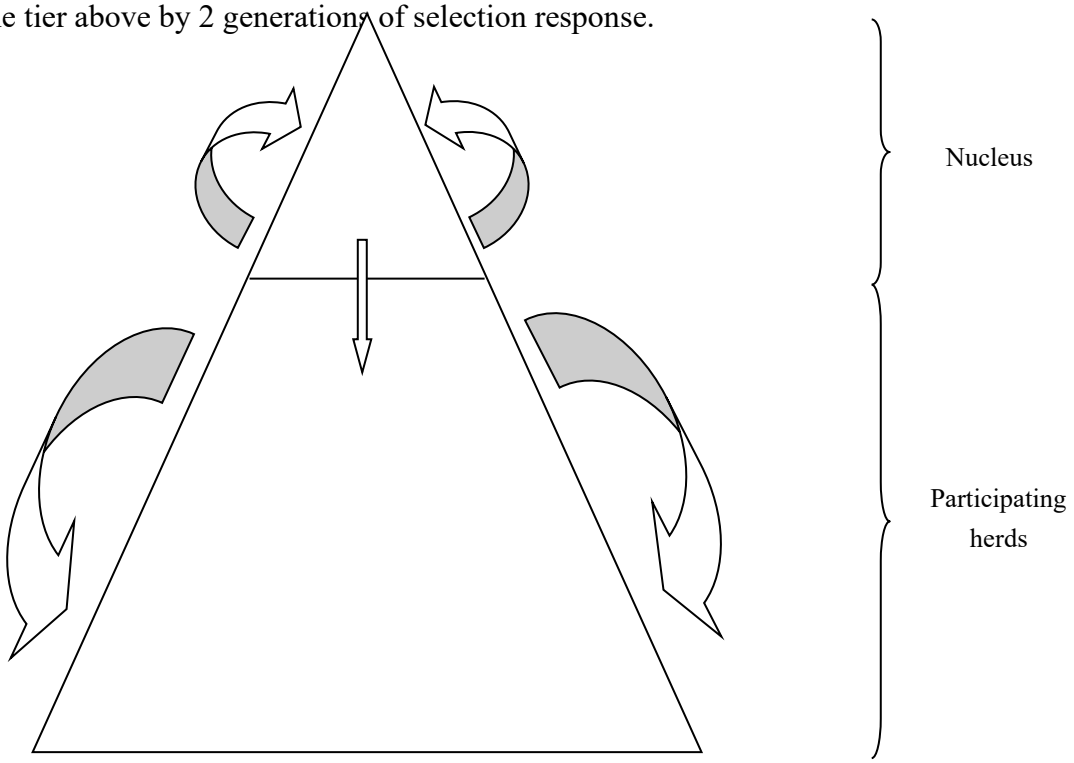
An industry can contain millions of animals. It is not worth including them all in a breeding programme due to measurement costs, recording costs, and lack of proper control. The solution is to concentrate efforts on relatively ***few breeding units (or nuclei)*** at the top of a pyramid structure and disseminate the superiority to the whole industry.

Closed nucleus breeding schemes

Closed nucleus schemes are closed in that no breeding stock are imported into the top level. In reality, there is usually a fair bit of migration between different flocks and herds in the top one

or two tiers. Closed schemes have evolved in most animal industries, driven largely by market forces. Here as some key properties of closed schemes:

1. Selection effort is only permanently effective in the nucleus - any temporary changes in lower tiers are diluted by importation from the nucleus
2. Nucleus breeding objective impact on the whole scheme
3. If lower tiers buy average males (an no females) from the tier above, they will lag behind the tier above by 2 generations of selection response.



Open nucleus breeding scheme

Nucleus breeding schemes were developed to circumvent the high costs arising from performance recording and selection. They do not require expensive infrastructure because recording is only done in the nucleus herd. Therefore they can be a good strategy for genetic improvement in non-industrialised countries which lack the expertise and structure required for operating an efficient genetic improvement programme based on artificial insemination (AI) and recording in the whole population. The best males are kept for breeding in the nucleus while the remaining selected males are used for breeding in the co-operating herds.

Open nucleus systems provide approximately 10% more genetic gain than a closed system because there are more animals which are potential candidates for selection. Such a system will integrate farmers' resources, reduce overhead costs and encourage more farmer participation.

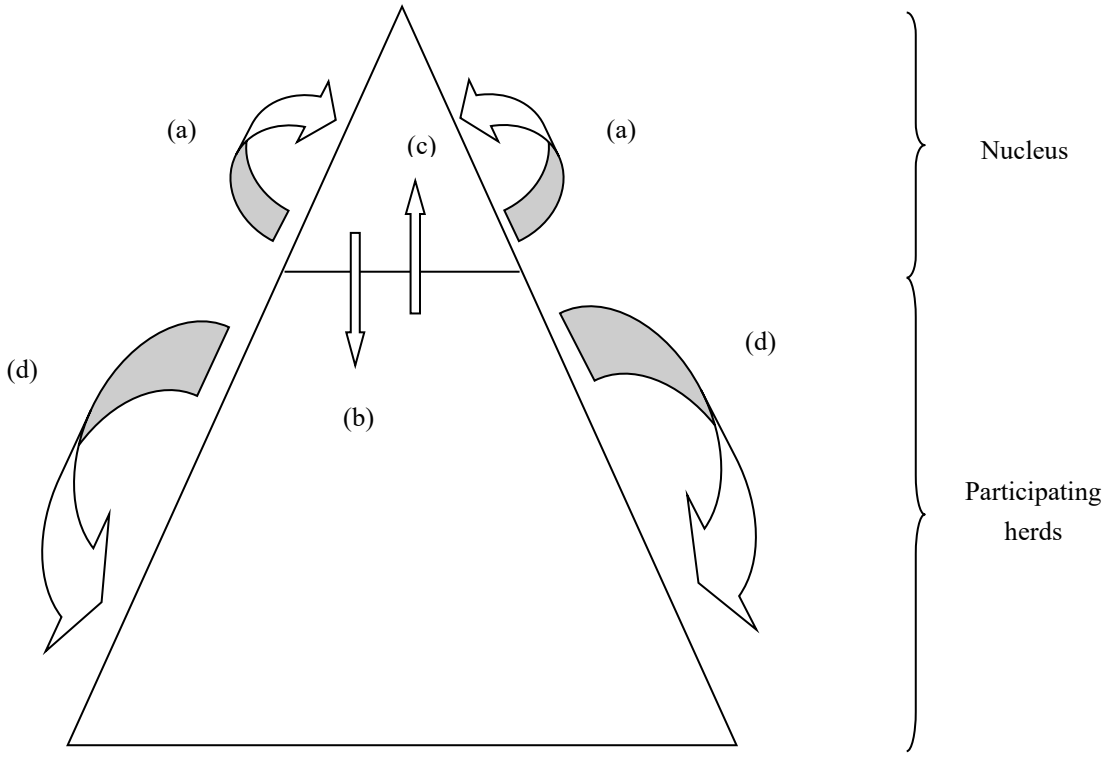


Figure 12. ***Two-tier open nucleus breeding structure with selection in nucleus and possibly in the participating herds, and downward and upward migration of genetic material.***

In an open nucleus breeding system (Figure 12):

(a) The nucleus is the tier that generates genetic gain and where sire selection is the main activity.

(b) There is movement of bulls from the nucleus to sire progeny in the field populations.

(c) In the nucleus, there is introduction of dams born in the participating herds which are selected subjectively based (depending on the breeding objective) on easily and cheaply measured traits.

(d) There is no selection in the participating herds. The bulls born in the nucleus are used to produce cows and to sire bull calves that are kept for use to mate a proportion of the cows in this sector. The dams in the participating herds are used to produce both bulls and cows in this sector.

Geographically diffused nucleus schemes

We can enjoy the full benefits of an open nucleus scheme without nominating one flock or herd to be the nucleus. We can create the elite 'nucleus' matings in the herds of birth of the female partners, with migration of semen to these herds. This relies on good pedigree information,

without which we lack the useful information about the tier of birth that a simple open nucleus scheme manages to exploit.

Genetic design of a breeding programme

1. *Development of breeding objectives*

The first phase in the genetic design of a breeding programme is that of developing the breeding objective.

Recall: Breeding objectives comprise those traits, which one attempts to improve genetically because they influence returns and costs to the producer.

Development of these objectives is made up of the following four phases:

(a) Specification of the production and marketing system;

(b) Identification of sources of income and expense;

(c) Determination of biological traits influencing revenue and cost;

(d) Calculation of the economic value of each trait in the breeding objective.

Breeding objectives should

- Incorporate the needs and economic interests of the producer
- Incorporate the ecological conditions under which animals will be raised
- Define the animal most suitable for the livestock producer under given production conditions.
- The choice of traits in the breeding objective can only be made in close consultation with the farmers, taking into account their needs

a) Specification of Production and marketing system.

In any animal production system's characterisation exercise, attempts should be made to collect information using simple procedures if the results are to be utilised fully for designing genetic improvement programmes.

E.g. Beef industry, Dairy Sector, Small-Ruminant production, Poultry production, Camel production (in Kenya)

Very little economic evaluation of breeding objectives has been attempted in the tropics. Breeding objectives should **account for inputs**, such as feed, husbandry and marketing costs, **as well as for outputs**, such as income from sale of milk, surplus offspring and cows.
These are difficult to quantify under most tropical conditions. Farmers are able to identify important attributes in their animals, however, actual economic data is difficult to come by.

b) Identification of sources of revenue and cost.

The development of a profit function for a production system requires identification of the actual sources of revenue and costs. Profit (P) is a function of revenue (R) and cost (C).

$$P = R - C$$

R and C should be derived on a cow-year basis as functions of all items contributing to R or C.

Example

Sources of revenue and cost for the Kenyan zebu

Revenue (R) = (culled heifers x value per individual) + (bulls x value per individual) + (amount of milk (kg) x price of one kg of milk) + (cull-for-age cows x value per cow).

Costs (C) = calf feed costs + heifer feed costs + bull feed costs + cow feed costs + calf health costs + heifer health costs + bull health costs + cow health costs + heifer reproduction costs + cow reproduction costs + heifer labour costs + bull labour costs + cow labour costs + fixed costs.

Fixed costs are those incurred by the producer independent of the level of herd production. All the other costs are **variable costs** and are influenced by the level of herd production.

c) Determination of Biological traits influencing revenue and cost

e.g. The biological traits assumed to have some influence on revenue and costs in Zebu cattle are listed in the table below:

A suggested list of breeding goal traits in the improvement of the Kenyan zebu breeds.

Milk production	*Female fertility*	*Growth*	*Longevity and survival*
Lactation milk yield	*Age at first calving*	*Preweaning daily gain*	*Preweaning survival rate*
Lactation length	*Calving interval*	*Postweaning daily gain*	*Postweaning survival rate*
		Mature live weight	*Productive herdlife*

Other traits such as:

Resistance to disease - a biological trait, which has been identified as an attribute desired by farmers. However, its inclusion as a trait in the breeding objective requires some definition of ***easy-to-measure*** selection criteria genetically correlated with disease resistance (i.e. a suitable indicator trait), or identification of ***loci that will allow marker assisted selection***.

Survival to a certain age (in the presence of disease challenge) may be an indirect measure of resistance to disease.

In Dairy Cattle, Traits to be considered would include:

Milk yield kg	Survival
Milk Fat %	Temperament
Milk Protein	Overall type
Mastitis susceptibility (SCC)	Teat Placement
Milking speed	
Services per conception	

Note:

It is a little presumptuous to give a single objective/ goal for all dairy producers. Managers differ in terms of:

- Skills required for dairy business
- Preferences
- Short and long term targets

In other animal species:

(Ref Table Adapted from Taylor and Bogart for list of traits)

d) Derivation of economic values of traits in the breeding objectives.

The main aim of any producer is to ***maximise returns/profits*** from the enterprise, which should be more efficient relative to the enterprises of other producers.

Efficiency in a production system can be defined as:

- maximise profit (= revenue - cost)
- minimise cost per unit of product
- maximise the ratio of revenue : cost.

In defining breeding objectives, definition of efficiency function has to correspond with the individual livestock producer's interest, which is to maximise profit rather than to minimise cost per unit of product. However, given the diversity in production systems, generally both maximisation of profit and minimisation of costs per unit of product are important.

Economic values have to be derived by changing genetic merit of the trait, but allowing no simultaneous change in genetic merit of the other traits.

Methods for deriving economic values can be grouped into two categories

i) Bio-economic modelling (Normative approach-- data simulation) - enable modelling of future market conditions

ii) Positive approaches, which involve analysis of field data- uses current prices

In both approaches, a production function (also referred to as a **profit function**), that must include all economic traits, is required.

Generally, using the normative approach, economic values are derived by studying the behaviour of the system as a reaction to changes in levels of the endogenous elements that represent genetic merit of the animals. There are numerous possibilities for applying different prices and efficiency functions, levels and sizes of the production system. Box 2 describes the different methods used in deriving economic values.

Box 2

Methods of deriving economic values

Generally, there are two methods that are used in deriving economic values from profit function:

(a) by accounting for unit changes in returns (marginal returns) and of costs (marginal costs) arising from improvement of a trait (marginal profit method, also referred to as partial budgeting method); and

(b) by partial differentiation of the function with respect to the trait of interest.

2. Genetic and phenotypic parameters

Unbiased heritability estimates and accurate correlations (genetic and phenotypic) among performance traits in the breeding objective and the selection criteria are required in order to design selection programmes for livestock. Reliable estimates of genetic parameters and economic values would allow for development of optimal linear selection indices that are directly applicable to programmes selecting for improved performance.

Genetic and phenotypic parameter estimates normally apply directly to the specific population and environment from which the data were collected. Thus, estimates may vary from one population to another depending on gene frequency, previous selection, and environment and past history of the population.

Structural design of a breeding programme: general considerations

There are major limitations that must be recognised before any breeding programme is designed for any situation

E.g. the Kenyan zebu breeds.

Field recording and AI are normally not feasible in the areas where these breeds are predominant. Breeding programmes must therefore be built on alternative methods of recording, on different selection methods and on natural service.

Recording and selection would be more optimal in nucleus herds where a certain proportion of the dams born in the participating herds are brought to participate in an open nucleus breeding scheme. In this scheme, as a result of recording, selection and planned matings, superior genes could be passed to the field population via natural service bulls.

Steps in establishing a nucleus breeding scheme:

1. Introduce a nucleus on institutional or government cattle stations.
2. Open the nucleus by finding farmers who are willing to join the scheme.

- The specific needs and social circumstances, as well as constraints of such farmers must then be incorporated in the breeding objective.
- Use should be made of existing advisory services, and existing structures and means of knowledge transfer to further select farmers based on certain criteria. These may include:
  (a) property rights on land;
  (b) ease of access to farm (e.g. by car, motorcycle or other available means)
  (c) access to water source

(d) size of herd;

(e) ability to keep basic records, to follow prophylactic programmes and to supplement their animals during critical periods;

(f) willingness to allow their best animals to be introduced into the nucleus or to be used for elite matings with semen from the nucleus, and to contribute (or sell) the resulting progeny to the nucleus.

Note: Most farmers may be reluctant to give out their best animals. However, ***incentive schemes may include***:

(a) passing on to farmers profits accrued from the sale of milk from their animals;

(b) incorporating the animal in the nucleus on a temporary basis (and then returning it later when it is pregnant or accompanied by a heifer calf, having been mated to 'superior sire');

(c) leasing the animal to the nucleus (at an agreed price).(Disease prevention may be a problem)

Alternatively, farmers could retain their best animals and elite 'nucleus' matings could be done in the field population with migration of semen or sires from the nucleus to these herds. This would rely on good pedigree and performance information, which the farmer would be expected to keep under the supervision of the nucleus.

In most zebu dominated areas, herd sizes are generally small. Therefore, the scope for selection among replacements within farms is small. Consequently, the potential of opening the nucleus for replacements from these herds may be limited, especially when an individual farm unit is considered. To circumvent this problem, use can be made of village herds each comprised of several individual farm units with a considerable number of cows from which replacements can be selected. Such a village-level programme would be particularly useful given the communal use of breeding bulls.

A nucleus can also be established in the form of ***a private enterprise***, which can initially be funded by loans from financial institutions or governments, and/or farmers who are willing to join.
Selection in such an enterprise would be relevant because farmers will be involved in its operation or even in owning the stock, as in ***group breeding schemes or co-operatives***. Increased revenues from sale of genetic material would absorb the costs of such an enterprise. For the enterprise to sustain itself, a compromise would have to be reached on what would be the cost of this genetic material leaving the nucleus. This genetic material would be in the form of young, untested bulls.
Note:

Young bull schemes have lower costs and are simpler to run than old bull schemes that use progeny-tested bulls.

In an old bull scheme, only progeny tested (proven) bulls are used in the nucleus and participating herds. Experiences from a progeny-testing scheme in a herd of Sahiwal cattle in Kenya confirm that such schemes can be inefficient in practice (Rege and Wakhungu 1992).

In a young bull scheme, untested bulls are used to sire dams both in the nucleus and participating herds. Two types of young bull schemes need to be distinguished:

- One with semen storage or waiting bulls;
- One without these possibilities.

When AI is possible, the bulls are slaughtered after a limited amount of semen has been collected; the semen is stored until daughter records are available to select semen from the best bulls for planned matings. In the absence of AI, the bulls are maintained (waiting bulls) until daughter records are available and the inferior ones slaughter thereafter. The rate of gain in such schemes would suffer because it takes a long time for the bull's daughters to be recorded. The problems and costs of using deep frozen semen or of maintaining waiting bulls should also not be underestimated. A young bull system without semen storage or waiting bulls has successfully been applied in a programme to genetically improve the N'Dama breed in the Gambia (Dempfle and Jaitner 1999). The use of young, untested bulls for AI or natural mating in the participating herds leads to a substantial reduction in the time-lag required for dissemination of the genetic superiority of the nucleus to this sector.

Summary:

Forming nucleus breeding schemes involves:

1. Formation of a central nucleus herd or flock from the best females available from member's own herds. The co-operating farmers manage and own jointly the nucleus herd.
2. Comprehensive recording and rigorous selection for commercially important traits in the nucleus. Co-operating farmers generally have a common breeding objective, hence directional selection is possible
3. The best males produced in the nucleus are retained there for breeding, whilst the next best males are the main source of breeding males for the member's herds/flocks. In this way, genetic progress/ improvement from the nucleus herd is passed onto members through the improved males
4. Elite females/ female replacement stock may be selected within the nucleus only (closed nucleus) or individual farmers who have above average females can have their dams "promoted" to the nucleus (open nucleus)

Genetic advantages of nucleus breeding schemes

1. A faster rate of response to selection is achieved because more accurate breeding values are obtained and there is a higher intensity of selection in the nucleus herds.
2. There is a genetic "lift" in the nucleus flocks due to the initial selection on farm
3. Inbreeding is avoided because a wider genetic pool is available for selection, hence the chances of genes in the population being identical by descent are reduced.
4. A wider pool of genes are available to individual farmers.

Other advantages of nucleus breeding schemes

1. By acting as a group, breeders can do thing not available to them as individuals—eg, buy superior rams
   --buy in genetic advice
   --use expensive technical services (AI or ET)

2. Continuity of selection objective is maintained—individual breeders not in a group tend to change their minds often—changing objectives in the group is often more rigid, hence progress is more directional
3. By acting as a group genetic progress is relatively "cheap"---the initial genetic lift is about 15% and annual gains of 2% can be obtained (one needs to remember that genetic gains are cumulative).

Sire referencing schemes

Introduction

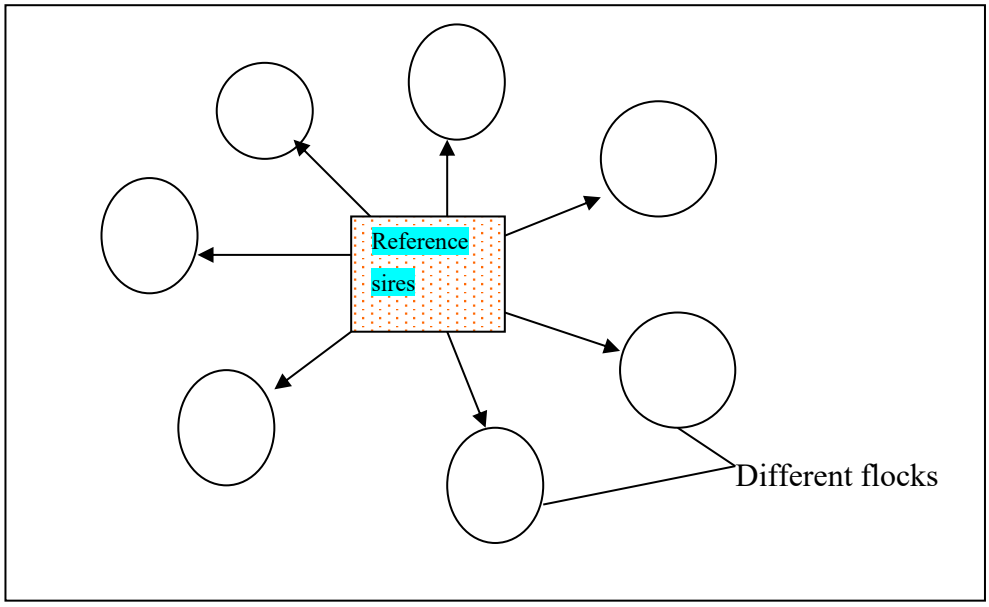
The problems with genetic improvement on individual farms include:

--small flocks / herds, hence few possible genetic comparisons between animals

--little genetic choice within the flock / herd

--unknown genetic merit from other flocks (in case of importation)—hence purchased animals may actually decrease the genetic worth of one's animals

Sire referencing scheme



How the scheme operates:

Assume that there are several farmers who own flocks that are isolated from each other genetically. The farmers may come together and select a group of sires that will then be centrally located and each producer / breeder will then chose to use a certain number of sires from the central location. Each of the flocks of participating farmers will have a proportion of its progeny from the reference sires, hence as a result, flocks will be genetically linked.

Animals on different farms can thus be effectively compared using BLUP procedures. Animals are compared within and across flocks. Use of reference sires is one way to increase the population size for genetic evaluation and improvement.

Benefits of sire referencing schemes

--they enable co-operation between breeders for their mutual benefit

--genetic comparisons of animals within the scheme are possible, and the rate of genetic gain within the population is increased because a higher selection intensity can be practised.

--there is a wider choice of animals, and animals with known estimated breeding values can be purchased.

--there is faster genetic improvement

In order for a sire referencing scheme to be effective, forward planning is essential. A calendar of planned events for the year should be prepared. For example in sheep production, a calendar of events may look as follows:

What happens	**When**
Members need to regularly meet, discuss and agree on breeding objectives	
Members record seasons of lambing	January
8 week weights are submitted and records checked for accuracy (pedigree records)	March
Visits on farm for scanning: within flock results are produced	June
Analysis of all records for the sire reference group is performed—deadline is negotiable –then run BLUP evaluation	June-July
Selection list of lambs, sires, dams (ranked in order of genetic merit) is sent to members	July
Members short list ram lambs as potential stock sires or reference sires	July
Ram lambs or stock sires are nominated to go forward for consideration as reference sires by other members	July
Ram selection day—vote on sires	July
Team of reference sires set to work	August
Summary reports – prepared and delivered	November
Various progress meetings of members are held throughout the year, and a final one in December	December

Design of dairy breeding programs

Dairy breeding programs are somewhat special case in the design of breeding programs as they have a 4-pathway structure. Also the dairy industry is relatively advanced in taking up new technologies, such as use of EBVs, A.I and other reproductive technologies. It is therefore an interesting case to study when it comes to the effect of new technologies on the breeding program design.

Worldwide dairy breeding is characterised by;

1. High degree of data recording, about 70% of commercial farms participate in milk recording schemes
2. Wide-spread use of A.I, high proportion of calves are born through A.I

The combination of widespread herd recording and A.I, i.e. using sires across many herds, provides a good structure for genetic evaluation. EBVs of both bulls and cows are comparable over different herds. Of all livestock industries, the use of information provided by EBVs is mostly accepted in dairy.

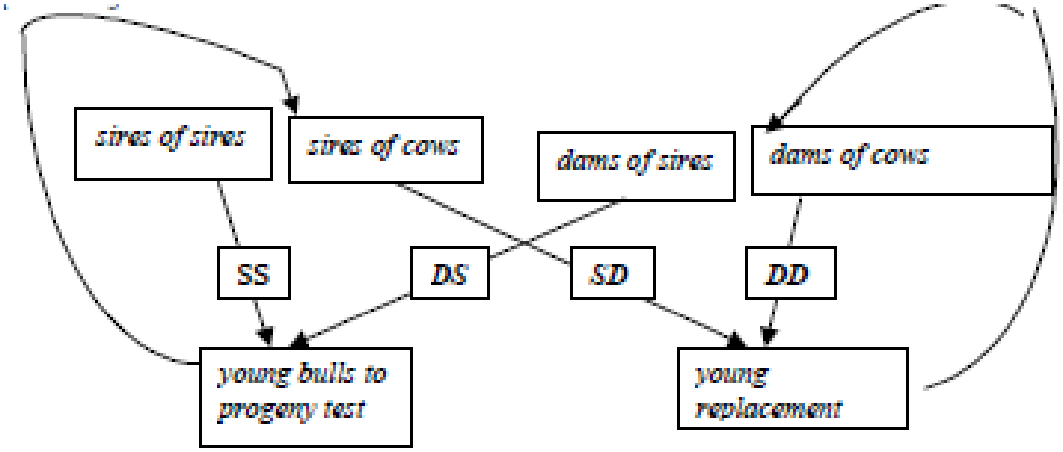
Progeny testing and the 4-pathway breeding structure

A typical design of a breeding program arises in the case of progeny testing: the 4-pathway breeding structure. The two main reasons for a special design is that

1. Many more females are needed for breeding than males
2. Males are selected based on a progeny test, as males do not produce themselves, but an accurate EBV is needed before they can be used widely.

Progeny testing is only a practical option if AI is used. AI facilitates testing of progeny across herds and the investment in progeny testing is only paid off if the bulls selected based on progeny tests can be widely used.

Progeny testing is expensive and obviously not all males born in the population are progeny tested. Only the best young males are progeny tested, i.e. the offspring of the very best dams and sires. The 4-pathway selection structure consists of the following pathways:



Elite matings to produce male calves to progeny test

- Sires of sires (SS): selection for elite sires
- Dams of sires (DS): selection for the elite cows

Elite matings to produce replacement females

- Sires of dams (SD): selection for the better sires
- Dams of dams (DD): selection for the better cows

The example shows some different parameters that need to be evaluated in deciding on a progeny-testing program. Those are;

1) The number of males progeny tested; and
2) The number of offspring generated per progeny tested male.

An important feature when designing a progeny test is to distribute the progeny over different herds or locations. In genetic evaluation, the phenotype from the progeny is compared with the phenotypes of progeny *from other sires*. If the animals in a particular management group are all descending from one sire, there is no basis for comparing them with offspring from other sires.

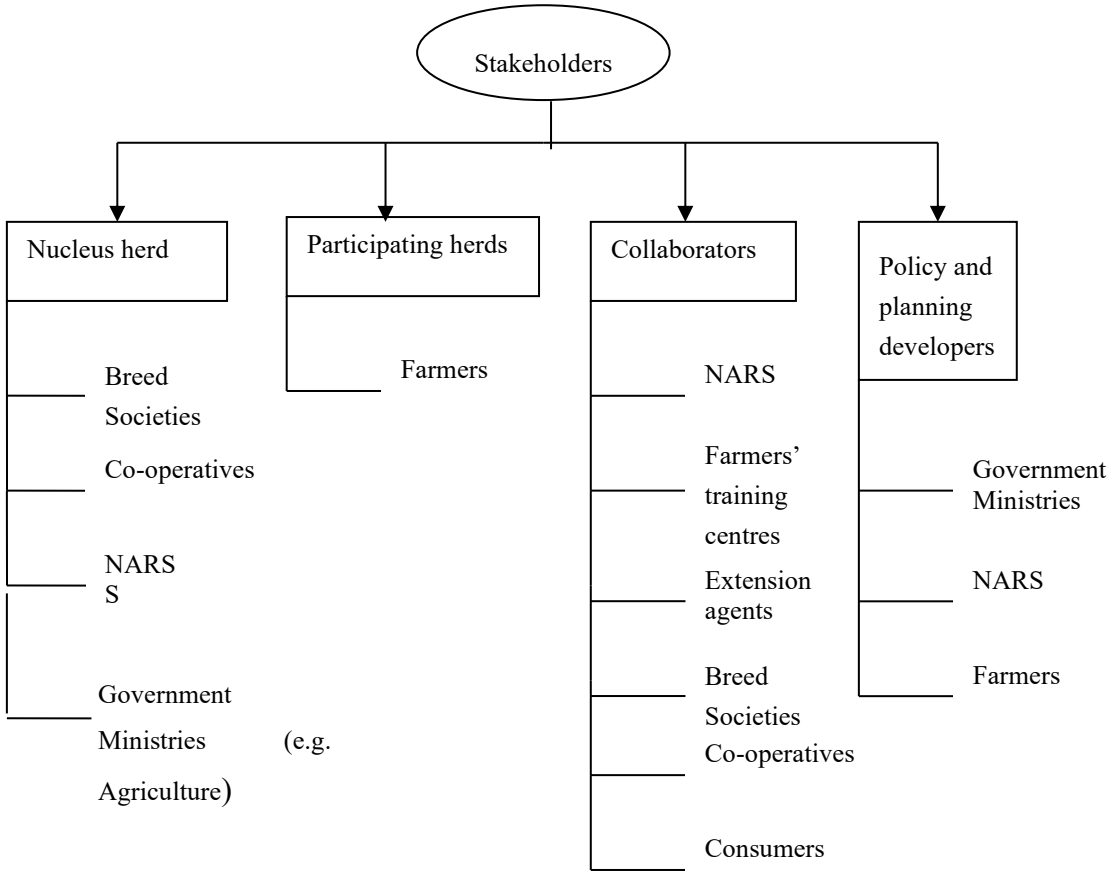
N/B: *The effective number of progeny is maximised when each progeny of a bull is compared with progeny from a maximum number of other bulls.*

Stakeholders and their potential roles

The number, type and roles of stakeholders vary depending on the stage of development of the breeding programme. In countries with well-developed and functioning breeding programmes, there may be many stakeholders each with well-defined roles.

An important aspect in starting up a breeding programme is ***identifying and defining roles of key players and how they interact.*** This is because activities related to the breeding programme have to take place at different locations but are interdependent. This requires interaction and communication between stakeholders.

Key Stakeholders can be classified into four broad groups:



Potential key stakeholders in a breeding programme. NARS = National Agricultural Research Systems e.g. universities, KARI and its research centres, research stations etc.

Note:

- Some subgroups are represented in more than one group.
- The nucleus herds can be owned by breed societies, farmer co-operatives or research stations owned by National Agricultural Research Systems (NARS). NARS as used here

include all public and parastatal organisations and private non-profit institutions, such as universities, that conduct research or work on the development or adaptation of technology, and policies that support agricultural and rural development.

- If a breed society owns the nucleus herd, then its overall role in the breeding programme would be the traditional role of the nucleus plus its role as a collaborator. Similarly, the overall roles of NARS would be equivalent to the summation of its roles as a collaborator, and policy and planning developer.

Nucleus herd

The nucleus' main **objective** is the improvement of the cattle.

- The nucleus should not be engaged in research *per se,* but should put in place collaborative mechanisms with appropriate stakeholders (e.g. NARS) to ensure that data coming out of the recording system are analysed, interpreted and the results used. Such partnerships will also facilitate identification of emerging researchable issues, which can support the effectiveness of the programme.
- The nucleus should also be involved in activities directly related to the cattle owners e.g. extension advice, open days, demonstrations and in identifying with the local community.
- Performance and pedigree recording, and selection should be the major preoccupation of the nucleus. The nucleus should, through its extension agents and scientists, screen all the cows joining it from the participating herds.
- Nucleus herds should provide superior genes to the participating herds and encourage farmers to purchase their products through advertisements (e.g. marketing of the breeding stock).

Participating herds (farmers)

Farmers own the animals and have responsibility for day-to-day decisions concerning the animals in their herds, i.e. feeding, health management etc. In an open nucleus system, this group may have the role of providing information on their best animals so that these animals can be introduced into the nucleus. These farmers are clients as well as proprietors of the breeding programme.

Collaborators

This group is important because it has the potential to provide ***additional human operational and management resources***. During the establishment stages, it is this group that could provide additional funding resources.

National Agricultural Research Systems (NARS).

These will be required in the early stages of development of a genetic improvement initiative (e.g. to contribute to capacity building) and when this initiative is operational. Qualified personnel to run such an initiative are likely to be found in these systems. Therefore, they are expected to be in charge of the technical support required in the implementation and running of the genetic improvement programme. Their roles could include:

- Developing the breeding programme and estimation of genetic parameters and economic values;
- Designing and evaluating the breeding programme (including quantifying extent of genetic progress);
- Electronic data processing and genetic evaluation;
- Designing a system of mating and exchange of breeding animals between locations/herds;
- Breeding advice;
- Training of staff to work in the breeding programme;
- Ensuring participation and co-operation through extension and education; and
- Research and development.

Farmers' training centres and extension agents.

- Farmers' training centres should train the farmers in aspects of animal production and veterinary sciences, and not just in genetic improvement.
- Farmers should be aware of what genetic change can achieve and how this can occur. Training of farmers is the responsibility of extension agents who should be able, with the help of researchers, to translate research information into simple terms for the farmer to understand.
- The extension agent is the contact person for the farmer in relation to improvement of animal production. Experience has shown that the level of management in a farm is positively correlated with the number of times a farmer is in contact with his/her extension agent.

Breed societies.

In a developed breeding programme, most of the roles of NARS outlined above are performed by breed societies. In such systems, the nucleus is normally made up of several herds located in different places and ***breed societies generally co-ordinate the breeding activities between locations.*** Their roles include:

- Registration of animals based on set breed standards;
- Maintaining national records for registered animals as deemed necessary, in the best interests of the breeders;
- Designing programmes to promote and protect the purity of the breed consistent with the set standards;
- Promoting interest, spreading knowledge and supporting responsible breeding;
- Educating members and the general public (e.g. through demonstrations) about proper care, maintenance and treatment of the breed; and
- Funding research work aimed at improving, protecting, preserving and promoting the breed.

Co-operatives.

Co-operatives can be farmer organisations when they are 100% owned and run by farmers. Low returns per unit of production, inaccessibility of resources and markets, and ignorance or lack of awareness are some of the problems farmers face. Through co-operatives, farmers create their own infrastructure of marketing and production support services e.g. services in animal breeding, health care, feed, other inputs and credit facilities.

Consumers.

- Drive the breeding programme in that they encourage the breeders and producers to focus the programme in ways that reflect the market (consumer) demand.
- Influence the breeding traits through their preferences and purchasing power. For example, if consumers prefer meat with less fat and are ready to pay high prices for lean meat, the breeding programmes must be adjusted accordingly to reflect these preferences.

Policy and planning developers

This group should ***create an enabling environment in responding to consumer and farmer needs.*** The government, through some of its departments, and with the help of NARS would have to act as a ***regulatory body*** by developing animal breeding policies that ensure that the breeding programme is consistent with the overall national goals. Such laws would regulate the activities of each stakeholder by clearly outlining the roles of each. The policies would have to cover a wide area.

This group should also provide the broad ***policy decisions required in planning, implementing and maintaining the operation***. It may be important to effectively involve farmers or farmer organisations in making these decisions. While farmers will not have the capacity to understand the technical aspects of the programme, they must be able to obtain practical interpretations associated with certain decision options.

Proposal for a breeding programme in Kenya

A breeding programme should be suited to local infrastructures. In some cases, this may mean compromising the ideal theoretical breeding plan. The following breeding programme is described as a basic model illustrating the possibilities for improvement of the indigenous Kenyan zebu breeds through selective breeding. The most important feature of the breeding programme is its feasibility to do something where previously little or nothing was achieved.

The programme would be located in an area where the Kenyan zebu breeds predominate. KARI's RRC at Kiboko in Makueni district could be the nucleus.

- A two-tier open nucleus breeding scheme supplying young bulls to the participating herds would be assumed.
- There would be intense recording of performance in the nucleus but none in the field population. Mating would be by natural service both in the nucleus and the participating herds.

Because of the paucity of information, this proposal is based on the following assumptions: (a) That (some sort of) breeding objective for the indigenous Kenyan zebu is developed taking into account improved productivity and conservation of genetic diversity;

(b) That economic values of the goal traits are available, and that these are relevant to the production and marketing conditions of the areas where the indigenous Kenya zebu breeds predominate;

(c)That reasonably reliable genetic parameters for the goal traits are available.

Consider that the breed serves a ***dual-purpose role*** producing both meat and milk.

Current production and marketing system

- milk is paid for on volume alone (not composition)
- payment for meat is based on weight, irrespective of the class of cattle.
- many aspects of carcass quality are currently not valued by the market.

The breeding goal traits that need to be considered are: ***lactation milk yield, age at first calving, preweaning daily gain, postweaning daily gain (to 18 months) and mature live weight.***

Cows from the participating village herds that join the nucleus during the foundation stages and thereafter, would be selected based on:

- milking ability
- the age at first calving,
- size,
- growth performance,
- conformation and condition.
- In some cases, test milking could be done.

The selection of these cows would be by technical experts (e.g. scientists) from the research station assisted by the cattle owner. Cattle owners normally know their animals and can provide the information needed to identify genetically superior animals. The animals transferred to the nucleus would be recorded there during the following year with the remainder of the nucleus herd and re-evaluated. This ensures that all cows in the nucleus are superior to the average of the field population.

Permanent environmental effects between herds would be accounted for by adjusting for effects, such as those of management and age of cows. Screening of the field population would include from various areas in the country.

The result would be a diverse population, but after several generations of breeding, it is expected that the animals would eventually be more ‘uniform’, reflecting the breeding objective.

A total population of 10,000 cows is envisaged and the size of the nucleus with performance recording would be 10% (1000 cows) of this population. A total of 200 villages located in these districts would participate. Each of these village herds would be expected to release their best cow to the nucleus. With a nucleus size of 1000 cows, progeny testing is not recommended. Therefore, selection in the nucleus would be based on individual performance tests and on performance of relatives. Thus, pedigree recording will be crucial in the nucleus. Progeny testing is feasible in large herds with several hundred cows. In this situation, where AI and semen storage facilities are absent, it would be risky to keep bulls alive while they are being progeny tested i.e. until their daughters have finished their first lactation. If a breeding programme is set at year 0, the first genetic benefits from progeny testing are not realised until after approximately 10 years. For this reason, pedigree selection is preferred.

In the participating herds, the bulls would be a community property. They would be in the care of a bull keeper who would be remunerated by the farmers through payments for natural matings. In cases where herd size is approximately 40 cows, one young bull would be supplied

for natural mating for each three-year period. In such cases, a bull keeper would not be required. Where the herds graze extensively, the bulls would be allowed to run freely with the cows, but each such group of village herds would be treated as a breeding unit. It would be the responsibility of the bull keeper to maintain records of services for payment and parentage. Bulls in natural service would be used for only three years and then replaced with young bulls from the nucleus.

Dairy Cattle Industry in Kenya

Dairy cattle production in Kenya is based on the rearing of exotic breeds, namely Holstein-Friesian, Ayrshire, Guernsey, Jersey and Brown Swiss, and their various crosses with each other and with indigenous breeds on either large scale farms (LSF), medium scale farms (MSF) or small scale farms (SSF). A national livestock census is necessary to obtain accurate population estimates, however, the country's total dairy population is estimated at 3.0 million (Peeler and Omore, 1997).

Dairy production in Kenya can be broadly viewed as an open nucleus breeding programme, however, two tiers would be evident, that of the "elite" breeders, and commercial producers.

Most of the dairy cattle are found in the highland districts of central, Rift valley and Eastern provinces. National milk production is about 2.0 billion litres/ yr produced as follows

Small scale producers = 60%

Large scale produces = 30%

Pastoralists = 10%

National average production is estimated at 1200-1500kg/cow/lactation. With most grade crosses averaging about 1900-kg/yr and some grade cows able to yield up to 21000kg/yr.

Several institutions have played an important role in the development and running of Dairy Cattle breeding in Kenya:

Breed Societies:

These were established in the early days of white settlers to safeguard the purity of various breeds. Each society maintained a herd book and set various standards for the herd-book register in order to promote the interests of specific breeders. In addition, breed societies were involved in organizing field days and livestock shows in order to popularize the breeds they represented.

The Kenya Stud Book (KSB):

The KSB was started in 1920 by a joint effort of the various breed societies. The objective of the KSB was to register various livestock for each breed society in an organized and centralized manner. This would include maintaining the pedigree herd registers, and running an upgrading scheme for all livestock breeds. The register was also to be a point of reference for sales and

movement of livestock within and outside the national borders. The KSB was run by representatives from each breed society.
After independence, the KSB was run by the government, under the ministry of Agriculture and Livestock development.

The Central Artificial Insemination Station (CAIS)
Set up in 1946, the mandate of CAIS as a semen production and distribution centre. CAIS ran a contract mating scheme for recruitment of bulls or bull calves from various sources. The centre also ran a progeny testing scheme for genetic evaluation of bulls used for Artificial Insemination (AI) country-wide.

AI Services:
AI started in 1935 among settler farmers.
In 1966, the government set up the Kenya National AI Services (KNAIS) to supply AI to all categories of farmers nationwide. KNAIS was also to ensure semen from the CAIS was used adequately in the progeny testing programme, and in sire selection for genetic improvement.

Milk Recording:
The first official milk recording scheme was started voluntarily by commercial dairy farmers in East Africa in 1949. It was then called the East African Milk Recording Service (EAMRS). This scheme was originally operated among large scale farmers who dominated the dairy industry then. The objective of the scheme was to generate data for herd improvement, management, research, training and planning.
The organisation collapsed in 1970. The Kenya government then set up the Kenya Milk Records under the MOALD.

The Kenya Milk Records (KMR)
KMR maintained the original objectives of the EAMRS. However, dairy farming was shifting from entirely Large scale to more medium and small scale.
Initially milk yields of individual cows were recorded daily on official milk sheets by the farmers then submitted to the central office weekly. Official recorders form the offices would visit the farmers bi-monthly to obtain milk samples for butter fat testing, and to check that submitted records were genuine.
The feedback provided to the farmers was:
- Lactation certificates for lactations greater than 180 days

- Herd production averages and butter-fat test results
- Breed averages

(KMR obtained a peak of 22,696 animals registered from 292 farmers)

1980's -- KMR had financial and management problems

1992 -- The government relinquished its control of the dairy industry

1994 -- KMR dissolved from GOK. Services offered by KMR and KSB were taken over by livestock producers who established the Kenya Livestock Breeders Committee (later Kenya Livestock Breeders Organisation).

Changed name from KMR to Dairy Recording Services of Kenya (DRSK) whose objectives were:

- Revitalise national milk recording
- Increase the number of farmers participating in milk recording
- Increase the flexibility of the milk recording scheme to enable small scale farmers to participate.

The Livestock Recording Centre (LRC)

The LRC was set up in 1974 as a joint venture between the GOK and Germany. It is a section of the animal production division of the Ministry of Agriculture, Livestock Development and Marketing. Its main objective was

- to collate data from KMR
- Estimate breeding values for AI bulls and provide the information to the CAIS so that the results could be efficiently used in the breeding programme.

The LRC operated effectively until 1982.

From Past to Present

Over time however, lack of co-ordination among CAIS, KSB, KMR and LRC jeopardised prospects of running a progressive breeding programme. Due to set-backs in processing records at both KSB and KMR, LRC was unable to carry out timely animal evaluation. This meant that the CAIS had no national performance base to determine what criteria to use in selecting breeding bulls for the country. Assessment was thus carried out based on farmer "know-how," leading to erratic fluctuations in genetic progress over time. Producers that were involved in the progeny testing programme run by CAIS lost confidence and were reluctant to have their animals used to test young sires.

Provision of AI services to farmers also changed. From a virtually free service, farmers now had to pay for AI. Initially this was met with a lot of reluctance, however, over time, private

companies entered the market and are selling semen from different countries all over the world. These include, World Wide Sires, ABS, CRI. Their services are appreciated because foreign sires come with genetic evaluation information, giving an indication of the potential productivity of the daughters of various bulls--a service which the CAIS is unable to provide. Prices for semen vary from as little as kshs 300 to Kshs 4,500 per insemination. Most commercial dairy farmers in Kenya no longer depend on the Central Artificial Insemination Station (C.A.I.S) for their supply of semen opting to import directly from any part of the world. The inability of the CAIS to provide genetic evaluation information has jeopardised the existence of a national strategic breeding program. Often times the imported semen is of bulls that have been selected depending on the selection objectives in their countries of origin. The selection goals in countries of origin are almost always different to those of the countries where this semen is eventually used.

The National Dairy Development Programme (NDDP), carried out jointly by the Kenyan government and the Netherlands government, was specially oriented to smallholder mixed farming in high potential densely populated areas and attempted to improve livestock productivity in this sector. A "zero-grazing" package was developed to solve constraints of land scarcity, overstocking and to improve milk production and cattle reproduction. To strengthen the work carried out by the NDDP, and to improve the link between research and extension, liaison was made with KARI (Kenya Agricultural Research Institution) and ILRI (International Livestock Research Institute). Records collected on small scale farms from these projects have been used in the economic analysis of the small scale farming system, and in assessing different aspects of animal nutrition and survival in the sector. When the project ended, on most farms collection of records also ended.

One result of all the changes in the industry is that **costs** of producing milk have been increasing. Producers are now beginning to look at efficiency of production rather than just volume of milk production. Improved efficiency is only feasible where pricing of products is fair, and marketing outlets are available.

Highly developed dairy industry is characterised by:

1. Use of high quality dairy stock
   -- High producing dairy breeds
2. Well organised production, processing and marketing system.

3. Surplus production with most of these countries satisfying the domestic market and leaving surplus for export
4. Application of modern milk production technology
   -- State of the art factories for processing
   -- High quality standards
5. Well developed disease control infrastructure of high quality laboratories and personnel. This is an important support service for the industry
6. Well organised recording systems, selection of animals based on national genetic evaluation results

Developed dairy Industry in Europe/N. America is of interest to Africa and other countries due to the following reasons:

1. As source of high quality genetic material for improving Africans breeding stock – sires, dams imported semen, embryo's.
2. Provision of technical know how and skills – setting the standards.
3. Provision of dairy products to other countries when there is a shortage.
   --Exports
   --Food relief packages or in case of epidemic

Constraints to improved Dairy Industry in Africa and E. Africa as a region

1. Prevalence of breeds with very low milk production potential
2. Diseases- high incidence of diseases some are endemic
3. Insect pests, Ticks, tsetse etc
4. Harsh climatic condition
   --Hot climate, intense solar radiation
   --Seasonal rainfall fluctuation
   80% of E. Africa is classified as ASAL
5. Lack of know how (skills in dairy)
6. Little or no selection and genetic evaluation of animals for improved milk production

Development of an organised dairy industry in e. African region will require the following:

1. Improvement in quality of livestock
   - Good dairy cattle – selection

- Crossbreeding—to upgrade indigenous animals and take advantage of their disease resistance
- Keeping of records and evaluation of animals based on performance
- Use of AI with knowledge of which sires are able to maintain high milk production in African environment

2. Proper disease control programmes
   -- Including vector control, controlled movement of animals
3. Provision of adequate high quality fodder and supplementary feeds
4. Efficient animal management
5. Availability of qualified manpower in areas such as:
   -- Animal Nutrition
   -- Animal Breeding
   -- Dairy technology
6. Well organised marketing
   - Infrastructure- efficient dairy co-operatives
   - Private farms
7. Improved infrastructure-Particular roads since milk is a highly perishable commodity and has to reach processing centres on time.
8. Incentives by Government particularly prices – give subsidy
9. Influence of capital by both public (Government) and private sectors/Banking awaiting of credit (terms)
10. Improved land tenure system

-- Free hold tittle of lease hold rather than Government ownership – stability/guarantee ownership facilitates investments and role.

A simple example

Table 19 shows, for a hypothetical example, the breeding goal traits together with their economic values, phenotypic standard deviation, heritability and the correlations between them. These parameters are based on the results of Kahi (2000) and, in that sense not solely hypothetical; however, Kahi's study was conducted in a production system very different from that under consideration. Therefore, economic values used here might change given the differences in the levels of inputs. Even so, some attempt at predicting genetic gain using this information may be instinctive. Selection is on an index based on these traits.

Table 19. *Economic values (v), phenotypic standard deviation (σ_p), heritability (h^2), genetic correlations (below the diagonal) and phenotypic correlations (above the diagonal) of the breeding goal traits.*

				Correlations				
Trait[a]	v (KSh)[b]	σ_p	h^2	MY	AFC	DG	PDG	LW
MY (kg)	18.9	229	0.12		-0.04	0.10	0.11	0.12
AFC (days)	-4.8	118	0.13	0.17		-0.25	-0.25	0.15
DG (g/day)	2.8	26	0.21	0.10	-0.25		-0.03	0.40
PDG (g/day)	9.1	30	0.26	0.10	-0.25	-0.03		0.47
LW (kg)	7.0	30	0.30	0.03	0.15	0.40	0.47	

[a]MY = lactation milk yield; AFC = age at first calving; DG = preweaning daily gain; PDG = postweaning daily gain; LW = mature live weight. Repeatability estimate of MY was assumed to be 0.55.

[b]KSh = Kenya shilling (KSh 1 = US$ 0.02 in December 1997).

The nucleus size is 1000 cows, 20% (200 cows) of which are screened from the commercial population. With a mating ratio of 1:40, a population of 1000 cows translates to a total of 25 bulls being required for natural mating. These bulls, from the age of two years, are used for breeding for three years. The generation interval is four years. Assuming a calving rate of 80% and that one-third of calving results in a bull calf that is suitable for breeding, this herd would produce approximately 266 bull calves each year. If 10 bulls are needed each year (i.e. 10/266), this gives a selection intensity (*i)* of 2.18. For the females, 50% are selected after the first lactation has been completed. This gives a selection intensity of 0.80. Assuming that one-third of the calves are progenies of first calvers (unselected cows), the effective *i* value equals 0.80 x 2/3 = 0.53. Since the restriction is that herd size should remain constant, culling intensity is maximised. Therefore, 32% of cows are culled due to age, and are replaced with females newly selected from within the nucleus and from the participating herds. For the females in the participating herds, one is selected from each village herd of 45 females (i.e. 1/45). The selection intensity is therefore 2.39. Selection is based on a test day milk yield, which is assumed to have a heritability estimate of 0.05 (lower than that of lactation milk yield in the nucleus). The generation interval from cows to their offspring is 6 years. Figure 15 shows an outline of the structure of this breeding programme.

As opposed to progeny testing where parents are specifically selected to breed offspring of a given sex, for pedigree selection only two selection pathways, sires to offspring and dams to offspring, are distinguished (Ruane and Smith 1989). However, in an open nucleus breeding scheme, a third selection pathway (that of commercial-born cows to offspring) should also be included. Table 20 shows the information sources for selection in these selection pathways.

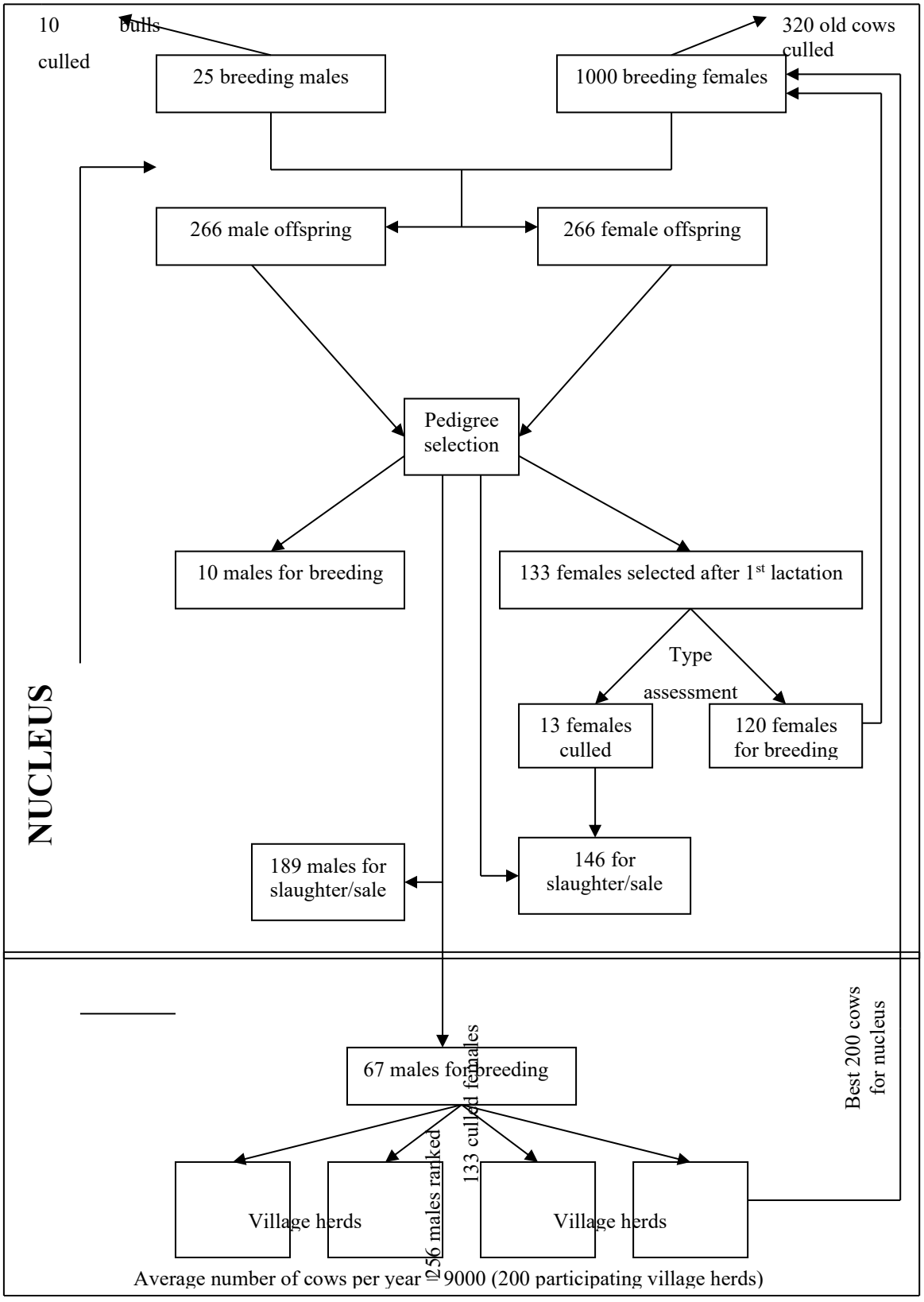




Figure 15. ***Outline of the structure of the breeding programme.***

Table 20. *The information sources for selection in the three pathways.*

Selection pathway	Information source[a]	No. records	No. calvings	Traits[b]
Bulls to offspring	Ind	1		DG, PDG
	S	1		DG, PDG, LW
	D	1	1	DG, PDG, MY, AFC
	DD, SD	1	3	LW, MY, AFC
	HSS	13	1	MY, AFC
	HSD	12	1	LW, MY, AFC
Nucleus-born dams to offspring	Ind	1	1	DG, PDG, MY, AFC
	S	1		DG, PDG, LW
	D	1	2	DG, PDG, MY, AFC
	DD	1	3	LW, MY, AFC

	SD	1	3	LW, MY
	HSS	13	1	MY, AFC
	HSD	12	1	MY, AFC
Commercial-born dams to offspring	Ind	1	1	Test day milk yield

[a]Ind = own performance; S = sire; D = dam; DD = dam's dam; SD = sire's dam; HSS = paternal half sib of sire; HSD = paternal half sib of dam.

[b]DG = preweaning daily gain; PDG = postweaning daily gain; LW = mature live weight; MY = milk yield; AFC = age at first calving.

The estimates of annual genetic gain in each breeding goal trait (Table 20) presented in Table 21 are based on the following:

$$\Delta G_k = \frac{\sum_l (i \times r_{IH} \times \sigma_A)_{kl}}{\sum_l L_l}$$

where: ΔG_k is predicted genetic gain per year for the kth trait over all selection pathways; *i* is the selection intensity; r_{IH} is the accuracy of selection; σ_A is the genetic standard deviation; and $\sum_l L_l$ is the summation of generation intervals in 'l' selection pathways.

Table 21. *Predicted annual genetic gain for the breeding goal traits.*

Breeding goal traits	Predicted annual genetic gain
Lactation milk yield (kg)	19.42
Age at first calving (days)	-0.18
Preweaning daily gain (g/day)	1.35
Postweaning daily gain (g/day)	2.72
Mature live weight (kg)	2.06

It should be noted that because of the deterministic approach, the results are completely determined by the input parameters (i.e. biological, technical and economic parameters). Therefore, the potential usefulness of this example is limited by the validity of the assumptions. These results show that with a single nucleus herd of 1000 cows, there can be genetic progress in these traits by simply selecting young bulls and cows, on the basis of pedigree and their individual performance – early growth performance for the bulls and first lactation performance for the females. Due to lower inbreeding and higher selection intensity, opening the nucleus seems to be the most appropriate means of increasing variance in a population, to levels above that in both nucleus and field population. Use is then made of the extra genetic variance by selecting cows born in the field population for use in the nucleus. Since the major questions in establishing breeding and recording programmes are those of return on investments, different breeding programmes need

to be compared in relation to the way they influence the accuracy of selection, genetic response, returns, costs and profit.

Apart from improving the animals in the field population, the described breeding programme will bring the participating farmers together and increase their contact with the extension agents. As already mentioned, contact with the extension agents will result in changes in animal management. With the support of the nucleus, participating farmers can come together and form a co-operative. Co-operatives normally work well when the membership is large and where incentives exist. In other words, co-operatives are money-driven. Money generated from the programme through sale of milk, and of culled and breeding animals should be enough to make the breeding programme self-sustaining. With time, these farmers could organise themselves to form a zebu cattle breeder's society which would then serve to set the breed standards and promote the breed as their own. Breeding programmes based on nucleus herds belonging to NARS should, in the long run, be controlled and owned by farmer organisations e.g. farmer co-operatives or breed societies.

Any efforts aimed at improving livestock production need to safeguard the functions of livestock at the individual farmer level as well as at the national level. It is important to note that quite often farmers are hesitant and slow in their acceptance of new ideas, as they have developed their own unique ways and methods of operation through experience. Any new ideas should thus have relevance to the farmers' problems and provide tangible solutions to these. The innovation should, whenever possible, carry enough evidence to suggest potential for success. Most farmers are prepared to believe and adopt change only after seeing the proposed change successfully demonstrated. Answers to constraints to production do not appear to lie in giving isolated attention to individual problems, but rather in a multidisciplinary approach that will have to recognise regional or zonal variations in rural life and even the hidden potential of the indigenous breeds.

PERFORMANCE RECORDING

A prerequisite to performance recording is proper identification of animals. The reason we record information on how animals perform, ***is so that we can identify and put a performance estimate (EBV or Estimated Breeding Value) on those parts of an animal (i.e. its genes), which are involved in transmitting production related performance to its progeny***.

What you see in an animal (its phenotype) is a combination of:

1. Its genotype (the sequence of genes that make up the animal).
2. A number, of what are known as, environmental influences such as grass, sunlight, wind, rain, trace elements, drenches, parasites, sickness and social stress expressing themselves as muscle, fat, bone etc.

How an animal performs will depend very much upon the extent to which it is exposed to these influences. For example, how much grass it eats, the number of parasites it is exposed to and how many drenches it receives or whether certain trace elements are absent in its diet. While all these influences are constantly changing and hence affecting the performance of the animal, the animal's genotype (genetic makeup) remains the same.

Note: *Animals do not pass on to their progeny the effects of the environment. Animals pass on their genes, which make up their genotype.*

Therefore in ***selecting an animal, in the absence of any performance information, you can quite easily be misled by its size, simply because the animal may have had access to the most grass/feed and not because it has the best genes for growth***. Effective performance recording involves keeping the influences of the environment the same for each animal while measuring its progress. This ensures that the difference in performance between animals can only be the result of the difference in their genetic makeup (genotype). With these genetic differences now isolated,

estimates of how the animals are likely to breed (EBVs) can be made. For successful design and implementation of breeding programs, performance records are also necessary.

Purposes of recording include;

1. Comparing performance for different breeding programs, managements.
2. Assisting in better management Levels of feeding (e.g. in dairy), determining whether the animals need treatment, changing feeds, weaning, mating, selling etc.
3. Economic evaluation and comparison (to optimise allocation and use of existing resources).
4. Recording performance of livestock is a prerequisite for livestock improvement.
5. Absence of records is a barrier to implementing and evaluating breeding schemes.

Those farms that participate and practice record keeping and utilise them have higher productivity than non-record keepers. They have increased genetic merit of their animals; increased value of their animals and produce, overall income for their farms is also increased. It is not only a matter of keeping records, but analysing and utilising records in farm decisions making.

What to record;

1. Those records essential to conduct improvement schemes and are cost effective.
2. Those records that can be maintained.
3. Those that can be analysed.

Taking records without using them (decisions about breeding, feeding and management) is a waste of time and money.

Records needed;

1. Most needed include those directly related to traits whose performance has been decided to be improved e.g. milk yield, body size, egg production, flock or litter size.

2. Some characteristics or traits that may affect production efficiency or costs and may change as the primary trait changes e.g. correlated traits.

Type of data to record;

1. Animal ID, herd ID, sample ID (Identification data).

2. Parents, pedigree (passport data).

3. Relevant parameter values.

4. Dates (ages calculated).

5. Weights (growth rates calculated).

6. Yields e.g. weekly lactation yield (lactation milk yield calculated).

Note: *Avoid hand calculations to avoid errors*

Example of standard operating procedures (SOP) for animal recording in Kenya by the Animal Breeding and Genomics Association of Kenya (KABGA) and the Kenya Dairy Farmers (KDFF)

I. Dairy

Introduction

Animal Breeding and Genomics Association of Kenya (KABGA) in executing its mandate, aims at establishing an effective and efficient animal recording system guided by internationally acceptable standards. **Animal recording is a process which involves; identification and registration of animals, recording of their performance, processing of the information, interpretation and dissemination of the results for decision making.** This SOP aims at providing recording guidelines that will help the farmers to achieve the objectives of increasing their income levels from their dairy enterprises, improved breeding stock, accurate and consistent information.

Identification

Identification is a process through which animals are identified using a combination of numbers and English alphabetical letters.

1. The animal should be identified after birth by applying an identification device.

2. The identification code should confer a unique identity to the animal within space and time. The Identification device constitutes;

- An ear tag applied on the left ear and/or
- A tattoo on the left ear and/or
- A radio frequency device in the rumen

3. Application of ID device in/on the animal is done by the farmer and/or an extension agent.

4. Generation of the unique identification codes is a preserve of KABGA

Identification codes

The identification codes carry the following information:

I. Country-254

II. County- constitute two numbers

III. Site code/Chilling plant code- constituting of two digits

IV. Animal number-will constitute three numerals

Management of the codes

1. At the beginning of September every year, KABGA shall announce the start of reception of herd sizes for preparation of identification codes for the subsequent year.

2. The farmer shall forward the number of the female animals on the farm and the expected birth rate to KABGA until end of October the year before the expected birth year.

3. The farmer shall apply for an identification device within Seven (7) days upon the animal giving birth.

4. KABGA shall generate the identification codes and prepare the identification devices. The device should be sent to the farmer Seven (7) days upon receipt of the application.

5. The device shall be applied on/in the animal immediately after receipt from KABGA.

Registration

1. All animals with ID devices shall be registered with the KDFF.

2. The information required for registration will be captured in an application for registration form. This will include the herd number, animal ID, its date of birth, sex, sire, grand sire, dam, grand dam and breed where applicable.

3. The application for registration will be filled by an agent from the extension department and submitted to the farmer KDFF within 14 days after identification of the animal

4. The KDFF shall register the animal and send the registration certificate to the farmer seven (7) days after receipt of the application form.

5. The Kenya Dairy Farmers Federation (KDFF) shall keep all the pedigree records (herd registry) and shall share it with KABGA as a pedigree file for evaluation.

Performance recording

1. The farmer shall notify KDFF of any birth immediately parturition occurs.

2. The KDFF shall on a monthly basis issue the farmer with a milk card that will have Site code, animal ID number, date of birth, dam/sire, date of calving, parity and input slots for daily milk yield

3. The farmer shall enter daily milk yield from the 10th day of lactation until drying.

4. Random inspection will be done by extension agents from the KDFF and KABGA to validate the data.

5. The completed milk card will be sent to the KDFF by the farmer immediately.

6. At the end of lactation the KDFF shall allow KABGA to access the data for analysis, generation of lactation certificate, and performance reports.

Lactation certificates

1. The certificate will be produced by KABGA within the first seven (7) days after completion of the lactation and submitted to KDFF for certification.

2. KDFF will surrender the certificates to the farmers through the chilling hubs or any other alternative, upon receipt of the requisite payments

3. The certificate will indicate the animal's parity, total lactation milk yield, 305-days Milk yield, average daily milk yield and Lactation length and graphical illustration of the performance.

Evaluation

1. KABGA shall perform periodic evaluation of animal performance and submit reports to the KDFF on a quarterly basis.

2. Genetic evaluation shall be carried out by the KABGA on an annual basis.

Performance reports

1. KABGA shall prepare and submit performance reports to the KDFF on a quarterly basis.

2. The report of the first quarter shall constitute performance parameters of the animals that completed their lactation within the quarter

3. The second quarter report shall be a half year performance report of animals that completed their lactation within the first half of the year

4. The third quarter report shall consist of animals that completed their lactation within the quarter

5. The final quarter report shall be an annual report on animal performance within the year.

6. All reports shall be submitted to the KDFF within 30 days after completion of the reporting period.

7. The genetic evaluation report shall be produced 30 days after release of the annual performance report.